

Internet Protocol (IP)

CCNA 200-301 V1.1

MODULE 4

Topics

4.1 IPv4 Basics

4.2 IP Packet and Interface Types

4.3 Binary Numbering System

4.4 Classful and Classless Addressing

4.5 IPv4 Subnetting

4.6 Subnetting in Other Octets

4.7 Supernetting

4.8 IPv6

4.9 Review



4.1 IPv4 Basics

CCNA 200-301 v1.1

Module 4

IP Address

Subnet Mask

Default Gateway

Public IP Addresses

Private IP Addresses

IP Address Assignment

Reserved IPv4 Addresses

Client IP Parameters

IP Address

Layer 3 address

IPv4

- 32 bit (binary digit) number
- Bits are organized in 4 octets (sets of 8)
- Usually translated to decimal for human convenience

IPv6

- 128 bit number
- Written in 8 hextets (4 hexadecimal numbers)

Address is logical

- Not burned into the hardware
- Can easily be changed or spoofed
- An interface can have more than one IP address as needed

Uniquely identifies a node on a network

- Two separate nodes cannot use the same IP address on the same network
- A node with a duplicate IP address will display an error and be unable to get on the network

```
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : hsd1.va.comcast.net
    IPv4 Address. . . . .           : 10.0.0.58
    Subnet Mask . . . . .           : 255.255.255.0
    Default Gateway . . . . .       : 10.0.0.1
```



Subnet Mask

Divides an IP address into two parts: network and host

- Same length as the IP address

A series of contiguous 1's that abruptly end, followed by 0's

- That dividing line is what divides the IP address into network | host
- Network ID falls under the 1's | Host ID falls under the 0's

Used by a node to make a decision:

- Is the destination on the same network as me?
- If so, ARP to find the destination MAC address and then send the packet to the destination
- If not, send the packet to the default gateway for further delivery

Subnet mask is required when you configure an IP address

Examples: 255.255.255.0, 255.255.0.0, 255.255.192.0, 255.0.0.0, 255.255.255.240



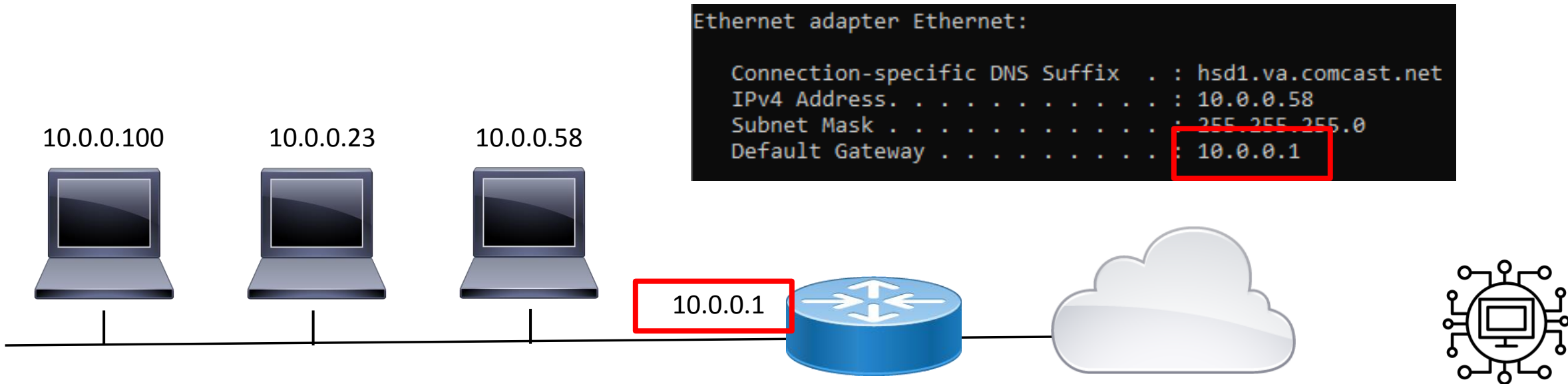
Default Gateway

The local router that connects hosts to other networks (usually the Internet)

- Hosts MUST have a default gateway to communicate with the outside world / other networks

Address is typically the first or last available IP address on that network

Examples: 192.168.10.1, 192.168.10.254



Public IP Addresses

An IP address that can be used on the Internet

Coordinated by the Internet Corporation for Assigned Names and Numbers (ICANN)

Regional Internet Registries (RIRs) assign region-specific blocks of IP addresses to ISPs

ISPs give these numbers to customers

- Usually via DHCP lease
- In some cases you can request/pay for static IP addresses

Examples:

- 18.134.5.6
- 192.169.3.77
- 207.45.128.32
- 172.32.1.2



Private IP Addresses

RFC1918

Used to extend the lifespan of IPv4 / conserves global addresses

Legitimate IP address, but not used on the Internet

- Can be used and routed on private, internal networks
- Cannot be directly connected to the Internet
- Addresses are hidden from the Internet; provides a level of protection

Private IP Address Blocks:

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

Must be NAT/PAT translated for traffic to travel on the Internet

- NATing is used to overcome the depleting number of IPv4 public addresses



IP Address Assignment

Method	Comment
Static	- Manually configured IP address and subnet mask in TCP/IP properties - Also known as “hard-coded”
DHCP	DHCP server on segment leases (temporarily assigns) IP address, subnet mask, and other options to requesting node
Self-assigned	- Node attempts to obtain a DHCP lease - Performed if no DHCP lease is available - Node self-assigns APIPA address 169.254.x.y - Windows and macOS are able to self-assign
No address	- No address configured/obtained - Device also cannot self-assign - Address is 0.0.0.0



Reserved IPv4 Addresses

Private

- Not used on the Internet
- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

APIPA

- Not routable
- Self-assigned
- 169.254.0.0/16

Loopback

- Not routable
- Assigned to a loopback interface
- 127.0.0.0/8 or a private address



Client IP Parameters

Most OSes will have both a command-line and GUI way of viewing IP address, mask, default gateway and DNS

- Commands will differ depending on the OS

Windows

- ipconfig
- Ipconfig /all
- Settings > Network and Internet > Status > (interface) Properties > IP Settings

macOS

- ifconfig
- ifconfig | grep inet
- scutil --dns | grep nameserver
- System Preferences > Network > TCP/IP

Linux

- ifconfig
- Ifconfig | grep inet
- ip route | grep default
- cat /etc/resolv.conf
- Ubuntu: Settings > Network > (interface gear icon) > Details



Windows 10 IP Configuration Example

```
C:\Users\chrys>ipconfig /all

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : hsd1.va.comcast.net
    Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
    Physical Address. . . . . : 74-04-F1-D2-D9-03
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2601:140:8500:d3d0::7723(Preferred)
    Lease Obtained. . . . . : Monday, August 5, 2024 3:17:29 PM
    Lease Expires . . . . . : Monday, August 12, 2024 6:26:52 AM
    IPv6 Address. . . . . : 2601:140:8500:d3d0:3419:a589:8f3f:66ee(Preferred)
    Temporary IPv6 Address. . . . . : 2601:140:8500:d3d0:adaf:88cb:b3ee:4e46(Preferred)
    Link-local IPv6 Address . . . . . : fe80::6a1e:4e66:3400:d60%7(Preferred)
    IPv4 Address. . . . . : 10.0.0.193(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Monday, August 5, 2024 3:17:28 PM
    Lease Expires . . . . . : Monday, August 12, 2024 1:38:11 PM
    Default Gateway . . . . . : fe80::966a:77ff:fe0c:3ef9%7
                                10.0.0.1
    DHCP Server . . . . . : 10.0.0.1
    DHCPv6 IAID . . . . . : 91489521
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-E1-4D-B7-5C-60-BA-60-5B-11
    DNS Servers . . . . . : 75.75.75.75
                                75.75.76.76
    NetBIOS over Tcpi. . . . . : Enabled
```



Windows 10 GUI Example

Properties

SSID:	Daisy WiFi
Protocol:	Wi-Fi 5 (802.11ac)
Security type:	WPA2-Personal
Network band:	5 GHz
Network channel:	157
Link speed (Receive/Transmit):	866/866 (Mbps)
IPv6 address:	2601:140:8500:d3d0::7723 2601:140:8500:d3d0:3419:a589:8f3f:66e e
Link-local IPv6 address:	fe80::6a1e:4e66:3400:d60%7
IPv4 address:	10.0.0.193
IPv4 DNS servers:	75.75.75.75 75.75.76.76
Manufacturer:	Intel Corporation
Description:	Intel(R) Wi-Fi 6 AX201 160MHz
Driver version:	22.250.1.2
Physical address (MAC):	74-04-F1-D2-D9-03



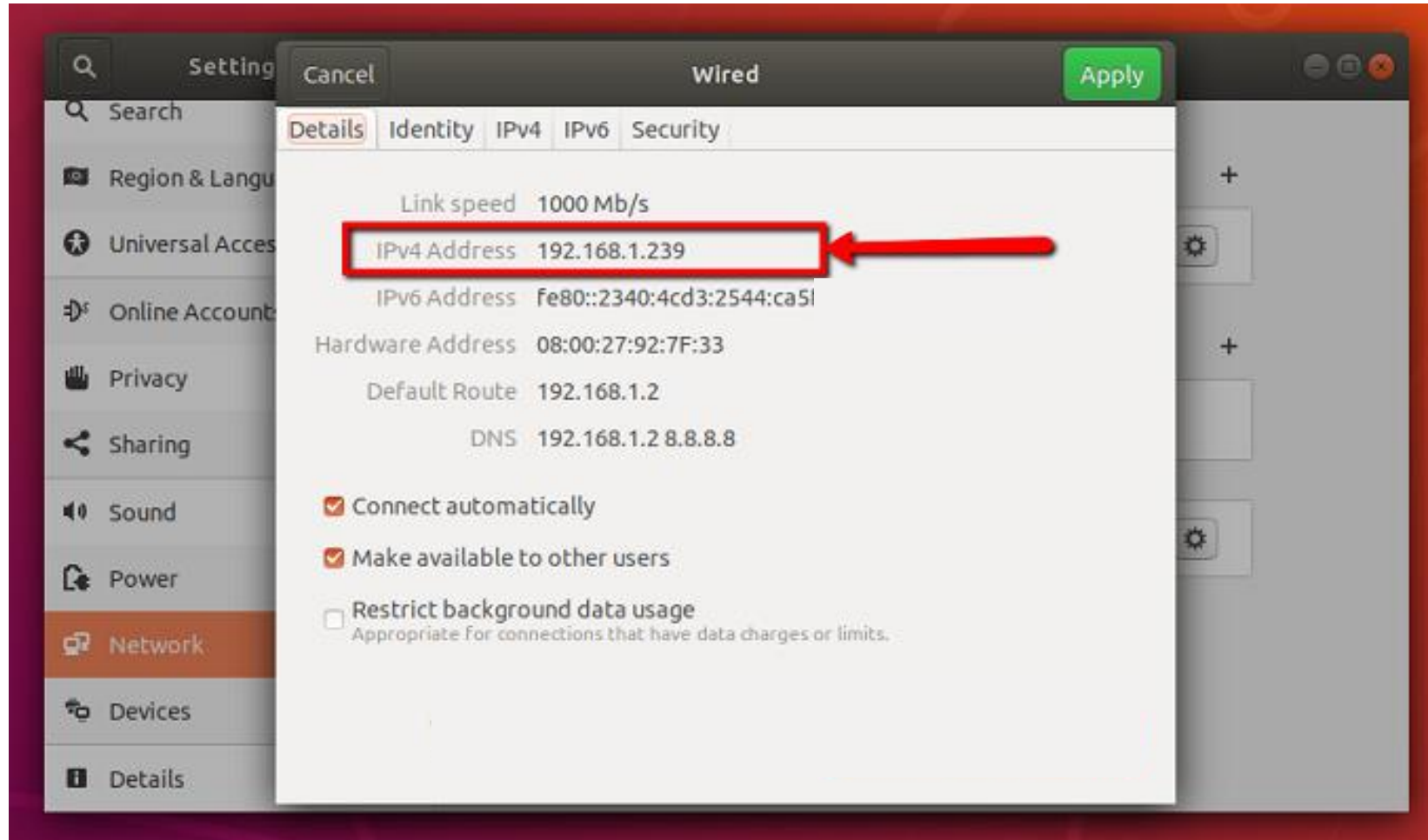
Linux IP Configuration Example

```
sara@sara-pnap:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 2000
    inet 10.0.2.15  netmask 255.0.0.0  broadcast 10.255.255.255
    inet6 fe80::c912:d4bf:3acb:f531  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:d6:51:a6  txqueuelen 1000  (Ethernet)
    RX packets 1442  bytes 632447 (632.4 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 1540  bytes 149029 (149.0 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 2012  bytes 180484 (180.4 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 2012  bytes 180484 (180.4 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```



Ubuntu GUI Example

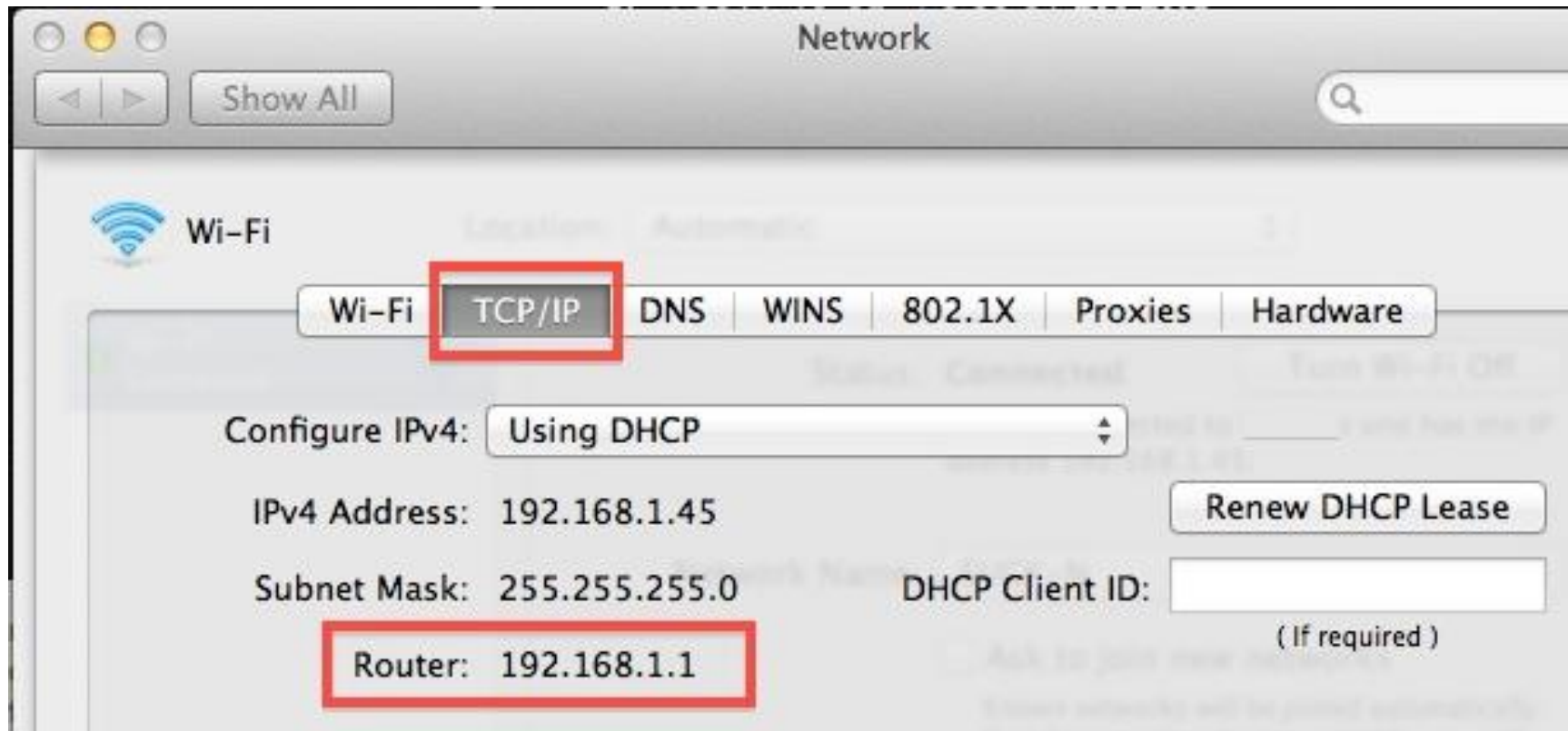


Mac OS X Configuration Example

```
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=3<RXCSUM,TXCSUM>
    inet6 ::1 prefixlen 128
    inet 127.0.0.1 netmask 0xff000000
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=1<PERFORMNUD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    ether 00:88:65:36:bc:8c
    inet6 fe80::288:65ff:fe36:bc8c%en0 prefixlen 64 scopeid 0x4
    inet 192.168.1.66 netmask 0xffffffff broadcast 192.168.1.255
    inet6 2602:306:3214:5aa0:288:65ff:fe36:bc8c prefixlen 64 autoconf
    inet6 2602:306:3214:5aa0:d54d:ccb2:6a7b:1f8f prefixlen 64 autoconf tempo
rary
    nd6 options=1<PERFORMNUD>
    media: autoselect
    status: active
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=60<TS04,TS06>
    ether 32:00:12:08:73:00
    media: autoselect <full-duplex>
    status: inactive
p2p0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> mtu 2304
```



Mac OS X GUI Example



4.2 IP Packet and Interface Types

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Module 4

Unicast

Multicast

Broadcast

Anycast

Link Local Address

Loopback

Virtual IP

Subinterfaces

Unicast

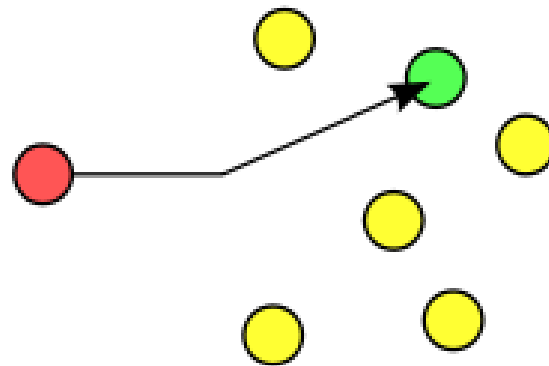
The most common type of IP packet

One sender → one receiver

A one-to-one transmission from one point in the network to another point

- Each point is identified by an address
- A Host will have both a Layer 2 and Layer 3 unicast address

In IPv6 a publicly routable unicast address is called a “Global Unicast address”



Multicast

One sender → multiple receivers

IPv4 address range = 224.0.0.0 – 239.255.255.255

IPv6 address range = ff00::/8

IP multicast is a bandwidth-conserving technology

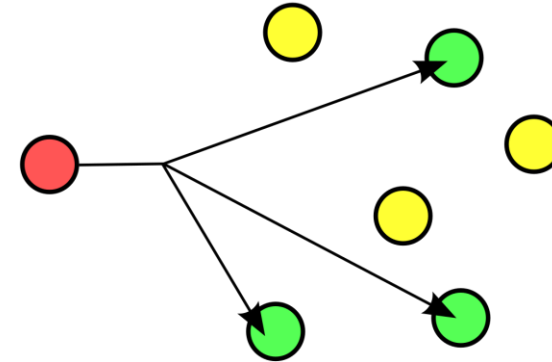
Reduces traffic by simultaneously delivering a single stream of information to potentially thousands of corporate recipients and homes

Receivers must “tune in” to the multicast as it’s being transmitted

Applications that take advantage of multicast include video conferencing, corporate communications, distance learning, and distribution of software, stock quotes, and news

IGMP is used to dynamically register individual hosts in a multicast group on a particular LAN

- Hosts identify group memberships by sending IGMP messages to their local multicast router



Broadcast

One sender → all receivers

Used when the destination address is not known

All nodes that “hear” a broadcast must process it

Global broadcast = 255.255.255.255

Directed broadcast = the host ID is set to all 1s

- IPv4 Examples:

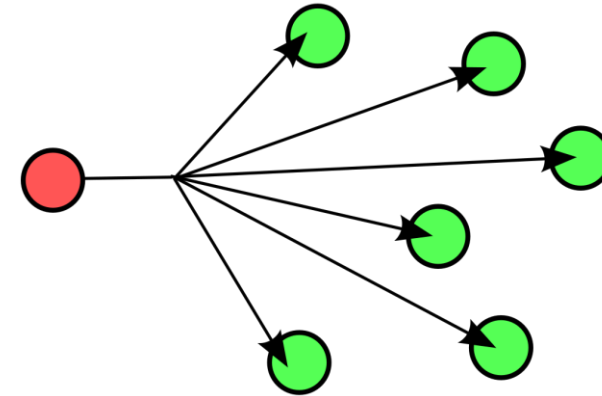
- 192.168.1.255
- 172.16.255.255
- 10.45.23.255

On a LAN, IP broadcasts will include a broadcast destination MAC address (FF-FF-FF-FF-FF-FF)

Many desktop operating systems prohibit the sending of Layer 3 broadcasts

- They will respond to a received broadcast

IPv6 does not use broadcasts



Anycast

A special type of unicast used by IPv6

Address looks syntactically like a unicast address

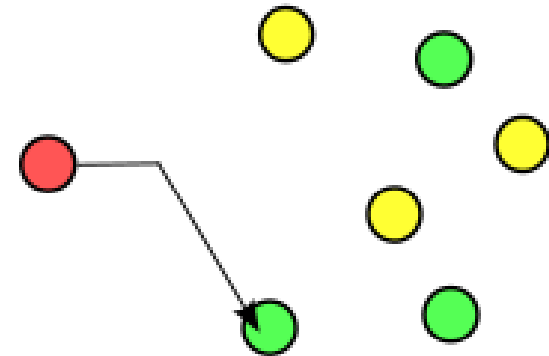
a single destination IP address is shared by devices (generally servers) in multiple locations. Routers direct packets addressed to this destination to the location nearest the sender, using their normal decision-making algorithms

6-to-4 tunneling is used to carry the anycast traffic on the Internet

Anycast routing is complex to set up

Widely used by content delivery networks and DNS

- Bring the content or service closer to end users.



Link Local Address

A secondary IPv6 address that is specific to the local network segment

Unicast, but not routable

You can ping it and communicate on your own segment with it

Self-assigned by the host

- A host will have a link local address even if it does not have a public (Global Unicast) IPv6 address

Basic structure = FE80:0000:0000:0000:abcd:abcd:abcd:abcd

Example:

fe80::2d1e:9ebb:92f0:d6e9%4

← Refers to the interface index (number)



Loopback

A software interface that can be used to emulate a physical interface

Loopback interfaces are always up and running and always available, even if other physical interfaces on the device are down

Loopbacks are often used on routers to identify the router and for diagnostics and testing

Default IPv4 loopback address is 127.0.0.1

Default IPv6 loopback address is expressed as ::1 (all 0s with a single 1)

It is pingable/reachable by other nodes if that node has a route to it

- Packet is sent to the address of a physical interface
- The router passes the packet internally to its loopback interface



Virtual IP (VIP)

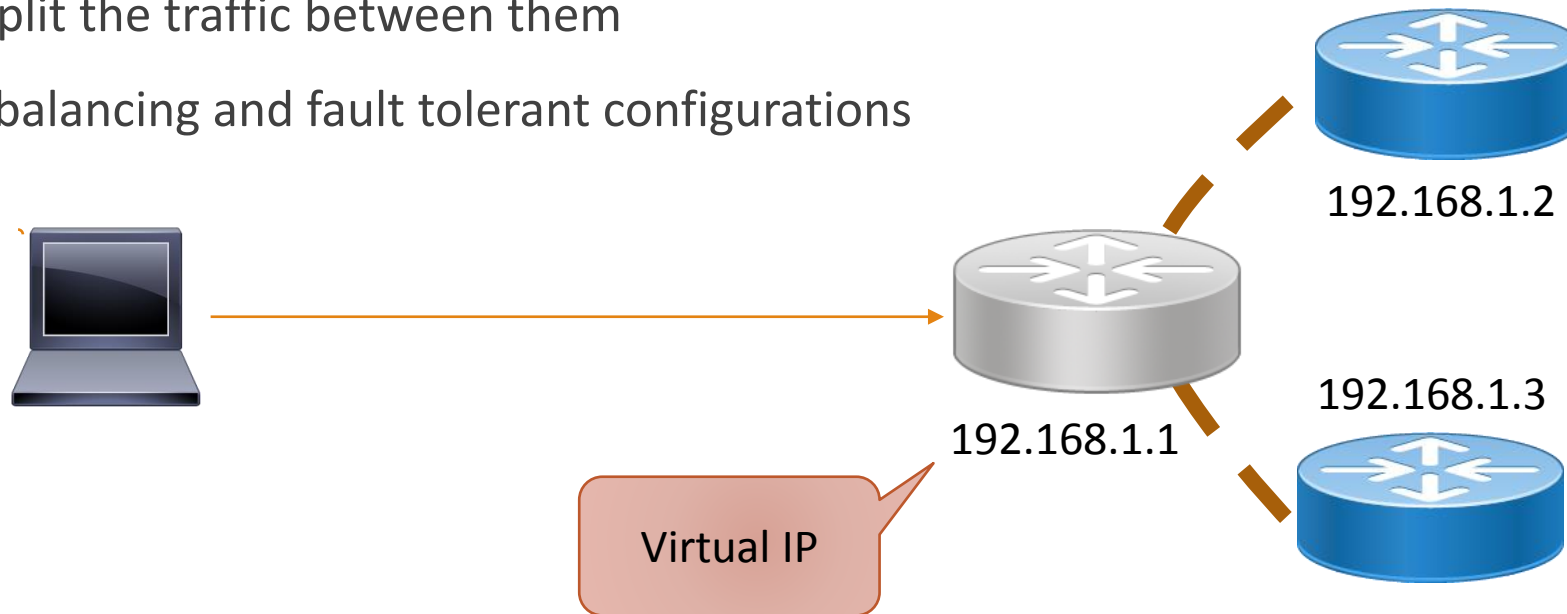
A IP address that is shared between two nodes

Each node has its own IP and MAC address

Client traffic is sent to the virtual IP

Nodes then split the traffic between them

Used in load balancing and fault tolerant configurations



Subinterfaces

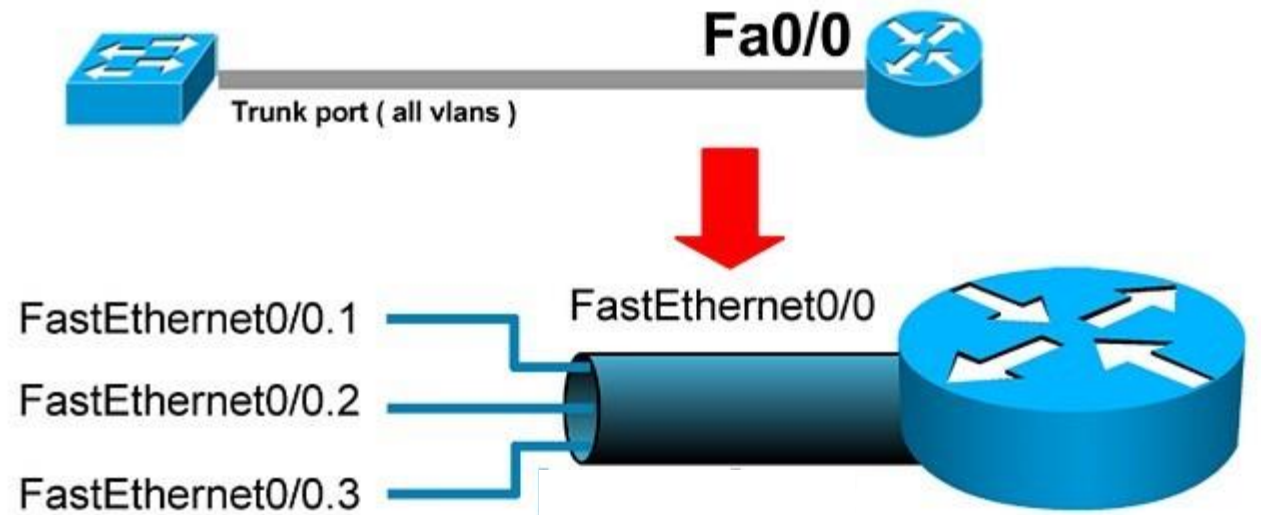
Used to create a “router on a stick” to route between VLANs

Logical interfaces are created on the router's physical interface

Each logical interface is the default gateway for its own subnet and VLAN

The physical interface itself will NOT be assigned an IP address

The physical interface must be 100mb (Fast Ethernet) or faster.



4.3 Binary Numbering System

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Module 4

Understanding Binary

IPv4 Addressing and Binary

Translating Binary to Decimal

Translating Decimal to Binary

Translating Hexadecimal to Binary

IPv4 Addressing Structure in Binary

Subnet Mask in Binary

Understanding Binary

Decimal

- Base 10 numbering system
- Used in every day life
- Numbers in a digit can be 0 – 9
- Moving a digit to the left multiplies by 10
- Moving a digit to the right divides by 10

Binary

- Base 2 numbering system
- Numbers in a digit can be 0 – 1
- Moving a digit to the left multiplies by 2
- Moving a digit to the right divides by 2
- Numbering system used in networking and computing



Counting in Binary

Decimal	Binary	Decimal	Binary
0	0	8	1000
1	1	9	1001
2	10	10	1010
3	11	11	1011
4	100	12	1100
5	101	13	1101
6	110	14	1110
7	111	15	1111
		16	10000



Counting in Binary

Decimal	Binary	Decimal	Binary
0	0	8	1000
1	1	9	1001
2	10	10	1010
3	11	11	1011
4	100	12	1100
5	101	13	1101
6	110	14	1110
7	111	15	1111
		16	10000



IPv4 Addressing and Binary

IP addresses are based on binary

IPv4 addresses are organized into 4 sets of 8 bits (octets)

Octets are separated by a dot (.)

1 st Octet	2 nd Octet	3 rd Octet	4 th Octet
			
11000000.10101010.00001111.00000011			



Translating Binary to Decimal

Translate the octets one at a time

If there is a binary 1 in a particular bit position, add the decimal equivalent

If there is a binary 0 in that position, do not add anything

128 64 32 16 8 4 2 1



Translating Binary to Decimal

Translate the octets one at a time

If there is a binary 1 in a particular bit position, add the decimal equivalent

If there is a binary 0 in that position, do not add anything

128	64	32	16	8	4	2	1
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>



Translating Binary to Decimal

Translate the octets one at a time

If there is a binary 1 in a particular bit position, add the decimal equivalent

If there is a binary 0 in that position, do not add anything

128	64	32	16	8	4	2	1
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>



Translating Binary to Decimal (cont'd)

Translate the octets one at a time

If there is a binary 1 in a particular bit position, add the decimal equivalent for that position

If there is a binary 0 in that position, do not add anything

$$128 + 64 + 0 + 0 + 8 + 0 + 2 + 0 = 202$$

1	1	0	0	1	0	1	0
_____	_____	_____	_____	_____	_____	_____	_____



Translating Decimal to Binary

Find the largest number in the octet – must be smaller than the number you are translating

Put a binary 1 in its position

202

Subtract from the total

Any digit where the remainder doesn't fit gets a zero

Repeat until nothing is left

128 64 32 16 8 4 2 1



Translating Decimal to Binary

Find the largest number in the octet – must be smaller than the number you are translating

Put a binary 1 in its position

202

Subtract from the total

- 128

Anything that doesn't fit gets a zero

74

Repeat until nothing is left

128 64 32 16 8 4 2 1

1



Translating Decimal to Binary

Find the largest number in the octet – must be smaller than the number you are translating

Put a binary 1 in its position

Subtract from the total

Anything that doesn't fit gets a zero

Repeat until nothing is left

$$\begin{array}{r} 74 \\ - 64 \\ \hline 10 \end{array}$$

128	64	32	16	8	4	2	1
1	1						



Translating Decimal to Binary

Find the largest number in the octet – must be smaller than the number you are translating

Put a binary 1 in its position

Subtract from the total

Anything that doesn't fit gets a zero

Repeat until nothing is left

$$\begin{array}{r} 10 \\ - 8 \\ \hline 2 \end{array}$$

128	64	32	16	8	4	2	1
1	1	0	0	1			



Translating Decimal to Binary

Find the largest number in the octet – must be smaller than the number you are translating

Put a binary 1 in its position

Subtract from the total

Anything that doesn't fit gets a zero

Repeat until nothing is left

$$\begin{array}{r} 2 \\ - 2 \\ \hline 0 \end{array}$$

128	64	32	16	8	4	2	1
1	1	0	0	1	0	1	0



Divide By Two

Another way to convert decimal to binary

1. Write down the decimal number
2. Divide the number by 2
3. Write the result underneath
4. Write the remainder on the right hand side as the least significant bit (smallest number)
 - this will be 0 or 1
5. Divide the result of the division by 2 and again write down the remainder
 - as the next least significant bit



Divide By Two (cont'd)

5. Continue dividing and writing down remainders until the result of the division is 0 – no fractions allowed!
6. Each remainder adds the next binary digit (going from right to left)
7. Go until you have nothing left



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

0

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

$101 / 2 = 50$ remainder $\rightarrow 1$

10

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

$101 / 2 = 50$ remainder $\rightarrow 1$

$50 / 2 = 25$ remainder $\rightarrow 0$

010

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

$101 / 2 = 50$ remainder $\rightarrow 1$

$50 / 2 = 25$ remainder $\rightarrow 0$

$25 / 2 = 12$ remainder $\rightarrow 1$ $\longrightarrow$ **1010**

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

$101 / 2 = 50$ remainder $\rightarrow 1$

$50 / 2 = 25$ remainder $\rightarrow 0$

$25 / 2 = 12$ remainder $\rightarrow 1$

$12 / 2 = 6$ remainder $\rightarrow 0$

01010

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

$101 / 2 = 50$ remainder $\rightarrow 1$

$50 / 2 = 25$ remainder $\rightarrow 0$

$25 / 2 = 12$ remainder $\rightarrow 1$

$12 / 2 = 6$ remainder $\rightarrow 0$

$6 / 2 = 3$ remainder $\rightarrow 0$

001010

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$ remainder $\rightarrow 0$ least significant bit

$101 / 2 = 50$ remainder $\rightarrow 1$

$50 / 2 = 25$ remainder $\rightarrow 0$

$25 / 2 = 12$ remainder $\rightarrow 1$ **1001010**

$12 / 2 = 6$ remainder $\rightarrow 0$

$6 / 2 = 3$ remainder $\rightarrow 0$

$3 / 2 = 1$ remainder $\rightarrow 1$

Go from right to left!



Divide By Two Example

No fractions allowed!

Starting number = 202

$202 / 2 = 101$	remainder $\rightarrow 0$	least significant bit
$101 / 2 = 50$	remainder $\rightarrow 1$	
$50 / 2 = 25$	remainder $\rightarrow 0$	
$25 / 2 = 12$	remainder $\rightarrow 1$	
$12 / 2 = 6$	remainder $\rightarrow 0$	
$6 / 2 = 3$	remainder $\rightarrow 0$	
$3 / 2 = 1$	remainder $\rightarrow 1$	
$1 / 2 =$ no way	remainder $\rightarrow 1$	most significant bit

11001010

Go from right to left!



Translating Hexadecimal to Binary

Hexadecimal = Base 16 numbering system

Each digit can be: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, or F

One hexadecimal number = 4 bits

- 0 = 0000
- 1 = 0001
- 2 = 0010
- 3 = 0011
- 4 = 0100
- 5 = 0101
- 6 = 0110
- 7 = 0111
- 8 = 1000
- 9 = 1001
- A = 1010
- B = 1011
- C = 1100
- D = 1101
- E = 1110
- F = 1111

Examples:

Binary	Hex
0000 0001	01
0001 0001	11
1000 1111	8F
1010 1011	AB



Things to Notice When Translating Binary

128	64	32	16	8	4	2	1		
0	0	0	0	0	0	0	0	0	Minimum number in an octet
1	0	0	0	0	0	0	0	128	
1	1	0	0	0	0	0	0	192	Add the next value
1	1	1	0	0	0	0	0	224	Keep adding the next value
1	1	1	1	0	0	0	0	240	Dividing line falls right in the middle
1	1	1	1	1	0	0	0	248	
1	1	1	1	1	1	0	0	252	Keep subtracting
1	1	1	1	1	1	1	0	254	Subtract from the maximum
1	1	1	1	1	1	1	1	255	Maximum number in an octet



Things to Notice When Translating Binary (cont'd)

128	64	32	16	8	4	2	1	
0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1	3
0	0	0	0	0	1	1	1	7
0	0	0	0	1	1	1	1	15 is just below 16
0	0	0	1	1	1	1	1	31 is just below 32
0	0	1	1	1	1	1	1	63 is just below 64
0	1	1	1	1	1	1	1	127 is just below 128
1	1	1	1	1	1	1	1	255

Stop Adding!
What is the next
number in the table?



Things to Notice When Translating Binary (cont'd)

128	64	32	16	8	4	2	1	
1	0	0	0	0	0	0	1	$128 + 1 = 129$
0	1	0	0	0	1	1	0	$64 + 4 + 2 = 70$
1	1	1	1	0	1	1	1	$255 - 8 = 247$
1	1	1	1	1	0	0	1	$255 - 4 - 2 = 249$
1	0	0	1	1	0	0	1	153
1	0	0	1	1	0	0	0	152

Add the 1s

Subtract the 0s

1 at end = odd

0 at end = even



IPv4 Addressing in Binary

Write out each octet

Separate the octets by a dot

192

127

11

4

11000000.10001000.00000011.00001000

224

128

15

64

11100001.01111111.00001111.01000001



IPv4 Addressing in Binary

128 64 32 16 8 4 2 1
— — — — — — — —

192 **127** **11** **4**
11000000.10001000.00000011.00001000

224 **128** **15** **64**
11100001.01111111.00001111.01000001

Do these all
look correct?
Does the
binary match
the decimal?



IPv4 Addressing in Binary

128 64 32 16 8 4 2 1
— — — — — — — —

192 127 11 4
11000000.10001000.00000001.100001000

224 128 15 64
11100001.01111111.00001111.01000001

Most of these
are wrong.



IPv4 Addressing in Binary

128 64 32 16 8 4 2 1
— — — — — — — —

192 **127** **11** **4**
11000000. 01111111.00001011.00000100

224 **128** **15** **64**
11100000.10000000.00001111.01000000

Now they are
right.



Subnet Mask in Binary

A subnet mask is also a 32-bit number

It is a series of binary ones that stop abruptly; the remaining digits are zeros

255	255	255	0	
11111111	.11111111	.11111111	.00000000	/24

255	255	0	0	
11111111	.11111111	.00000000	.00000000	/16

255	0	0	0	
11111111	.00000000	.00000000	.00000000	/8



Subnet Mask Dividing Line in Binary

The subnet mask divides the IP address into two parts: Network ID and Host ID

The dividing line is where the binary 1s end and the 0s begin

Subnet Mask	11111111.11111111.11111111.00000000
IP Address (binary)	11000000.10101000.00001010.00000001
IP Address (decimal)	192 . 168 . 10 . 1
	Network ID Host ID



Let's Talk About the Subnet Mask

The subnet mask can only be certain decimal values:

255, 252, 248, 240, 224, 196, 128, 0

Examples:

- 255.0.0.0 11111111.00000000.00000000.00000000
- 255.240.0.0 11111111.11110000.00000000.00000000
- 255.255.0.0 11111111.11111111.00000000.00000000
- 255.255.224.0 11111111.11111111.11100000.00000000
- 255.255.255.0 11111111.11111111.11111111.00000000
- 255.255.255.128 11111111.11111111.11111111.10000000
- 255.255.255.252 11111111.11111111.11111111.11111100

- You cannot intermix 0s and 1s
- The 1s must start at the far left
- Continue until they abruptly stop
- The remainder of the mask is 0s

4.4 Classful and Classless Addressing

CCNA 200-301 v1.1

Module 4

Classful Addressing

IP Address Classes

Classless Addressing

VLSM

CIDR Notation

Classful Addressing

RFC 791

The Internet's first major addressing scheme for IPv4

Five classes: A, B, C, D, E

Class A, B, and C (their entire address range) are sometimes called “major networks”

Classes distinguished by the high order bits

The Internet Assigned Numbers Authority (IANA) oversaw all classful network assignments

The Host portion was assigned by local organization's network administrator

Routers originally processed packets according to their classful network



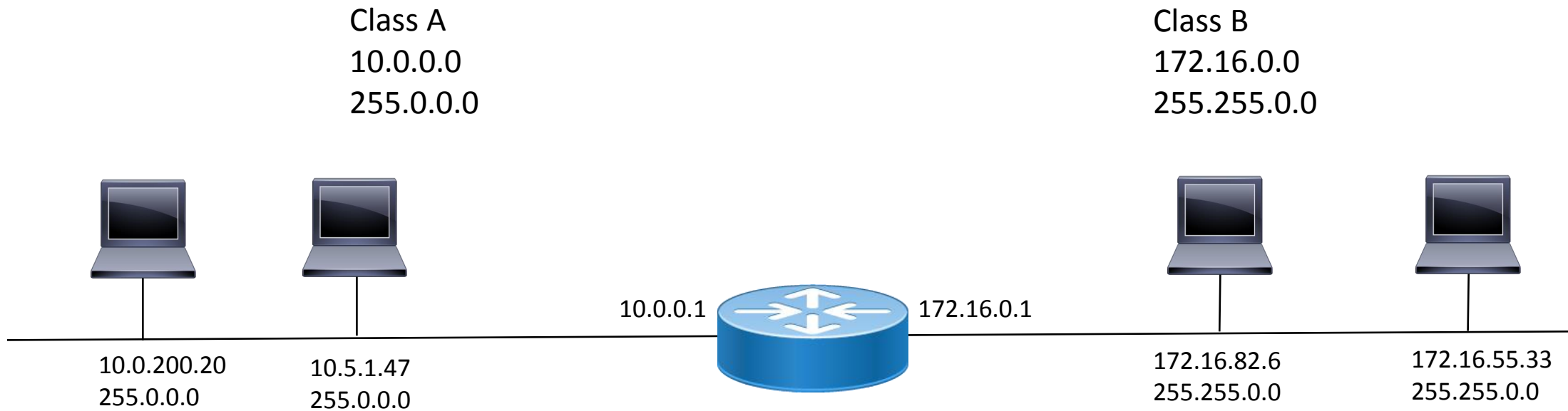
IP Address Classes

Determined by high order (far left / greater) bits in first octet

A -	0 0000000	0 - 127.x.y.z	255.0.0.0	/8	}	Major network
	0 1111111					
B -	10 000000	128 - 191.x.y.z	255.255.0.0	/16	}	Major network
	10 111111					
C -	110 00000	192 - 223.x.y.z	255.255.255.0	/24	}	Major network
	110 11111					
D -	1110 0000	224 - 239.x.y.z	No mask - multicasting			
	1110 1111					
E -	1111 0000	240 - 255.x.y.z	No mask - experimental			
	1111 1111					



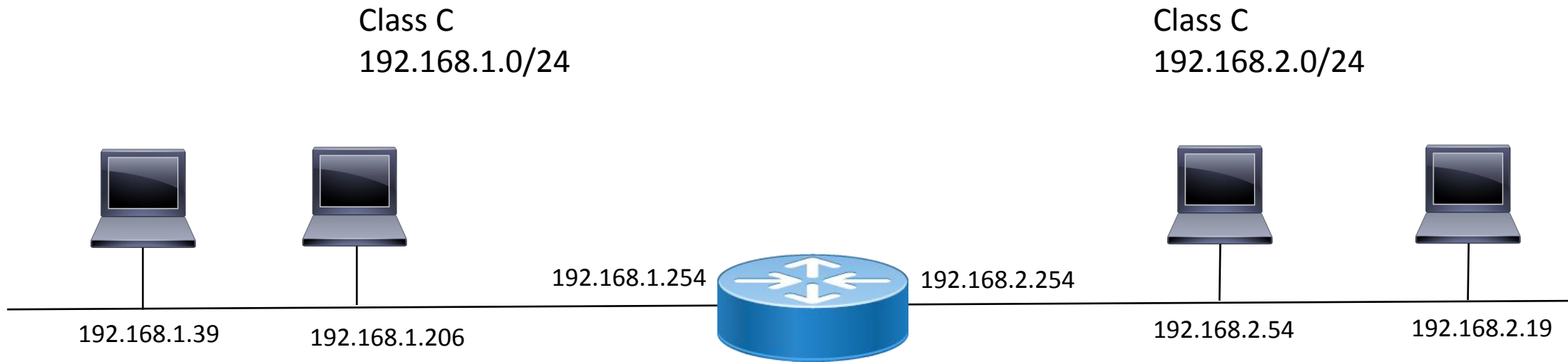
Classful Addressing Example #1



Major network boundary
Where two different address classes meet



Classful Addressing Example #2



NOT a major network boundary
Networks are from the same address class



What is Classless Addressing?

An attempt to reduce the wastefulness of classful addressing

Customize a default subnet mask to reduce the number of IP addresses

Used in subnetting to divide a classful network into smaller subnets

The “longer” the subnet mask gets (higher numbers it has), the fewer hosts in that range

255.0.0.0 → 255.255.0.0 → 255.255.255.0

shorter → longer → even longer

255.255.255.0 → 255.255.255.128 → 255.255.255.240 → 255.255.255.252

shorter → really long



Variable Length Subnet Mask (VLSM)

Customized (classless) subnet mask

Used to maximize IP address utilization in a network

Often more than one VLSM will be used in a larger network

Requires careful subnet design to be implemented well

VLSM Examples:

- 255.255.255.128
- 255.255.248.0
- 255.240.0.0



Understanding CIDR Notation



Classless Inter-Domain Routing (CIDR) Notation

CIDR is the use of VLSM on a network (particularly the Internet)

CIDR notation is a shorthand way of representing a subnet mask

- The notation can apply to any length subnet mask, classful or classless
- Created by counting the number of binary 1s in the subnet mask

Examples:

Dotted decimal	Binary	CIDR
◦ 255.255.255.128	11111111.11111111.11111111.10000000	/25
◦ 255.255.248.0	11111111.11111111.11111000.00000000	/21
◦ 255.240.0.0	11111111.11110000.00000000.00000000	/12



Working with CIDR

- Default subnet masks have the 1s end at a dot
- Classless subnet masks can have the 1s end between dots

255 = 11111111

/8 = 11111111.00000000.00000000.00000000

/16 = 11111111.11111111.00000000.00000000

/24 = 11111111.11111111.11111111.00000000



Working with CIDR (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

0

0

0

/8 = 11111111.00000000.00000000.00000000



Understanding CIDR Notation

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

128

0

0

/9 = 11111111.10000000.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

192

0

0

/10 = 11111111.11000000.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

224

0

0

/11 = 11111111.11100000.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

240

0

0

/12 = 11111111.11110000.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

248

0

0

/13 = 11111111.11111000.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

252

0

0

/14 = 11111111.11111100.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

254

0

0

/15 = 11111111.111111110.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

0

0

/16 = 11111111.11111111.00000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

128

0

/17 = 11111111.11111111.10000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

192

0

/18 = 11111111.11111111.11000000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

224

0

/19 = 11111111.11111111.11100000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

240

0

/20 = 11111111.11111111.11110000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

248

0

/21 = 11111111.11111111.11111000.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

252

0

/22 = 11111111.11111111.11111100.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

254

0

/23 = 11111111.11111111.11111110.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

0

/24 = 11111111.11111111.11111111.00000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

128

/25 = 11111111.11111111.11111111.10000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

192

/26 = 11111111.11111111.11111111.11000000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

224

/27 = 11111111.11111111.11111111.11100000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

240

/28 = 11111111.11111111.11111111.**1111**0000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

248

/29 = 11111111.11111111.11111111.11111000



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

252

/30 = 11111111.11111111.11111111.11111100



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

254

/31 = 11111111.11111111.11111111.11111110



Understanding CIDR Notation (cont'd)

Decimal
Conversion
Table

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
------------	-----------	-----------	-----------	----------	----------	----------	----------

255

255

255

255

/32 = 11111111.11111111.11111111.11111111



4.5 IPv4 Subnetting

CCNA 200-301 v1.1

Module 4

What is Subnetting

Classical Subnetting Formulas

Subnet Calculator Table

Delta Method

Verifying with Binary

Finding Subnet IDs

Supernetting

What is Subnetting?

Dividing a larger network into smaller sub-networks (subnets)

We typically use VLSM and classless addressing to create the subnets

As with classful networks, subnetworks are also connected by routers

All hosts on the same subnet (including the default gateway) must use the same subnet mask

- Required so that all can agree upon the range of addresses that belong to that subnet



Classical Subnetting Formulas

Calculate the # of **subnets**: 2^n

- where n is the number of bits you borrowed from the host side

Calculate the # of available **hosts** in a subnet: $2^n - 2$

- where n is the number of bits left for the hosts

You subtract 2 because:

- The very first address represents the entire subnet ID
- The very last address represents the subnet broadcast
- The subnet ID and broadcast addresses cannot be given to hosts

We have a better way than this ...



Subnet Calculator

Memorize this Tool!

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many
subnets you
need

How many bits
to move the
mask

“What if I only need 20 subnets?” You still need 5 bits because 4 won’t be enough.
You’ll have spares 😊

When subnetting (dividing a network) be sure to move the mask “down” to the right!



The Delta Subnetting Method

High Level Steps

1. Identify the starting subnet mask
2. Determine # of bits to move the mask
3. Move the mask and identify the delta
4. Use the delta to calculate subnet IDs and IP ranges



1. Identify the Starting Subnet Mask

192.168.1.0/24

Original subnet mask /24

192.168.1.

Network

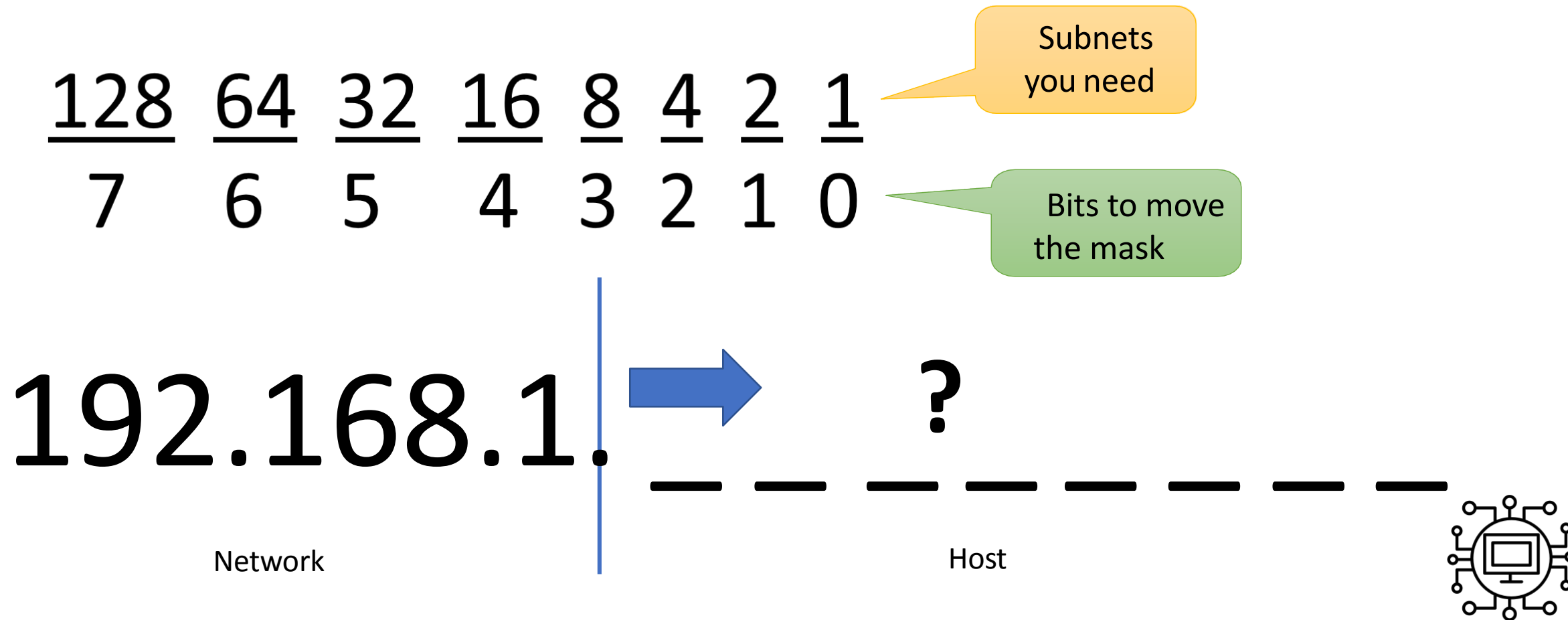
Break out last octet
into 8 binary bits

— — — — — — — —

Host

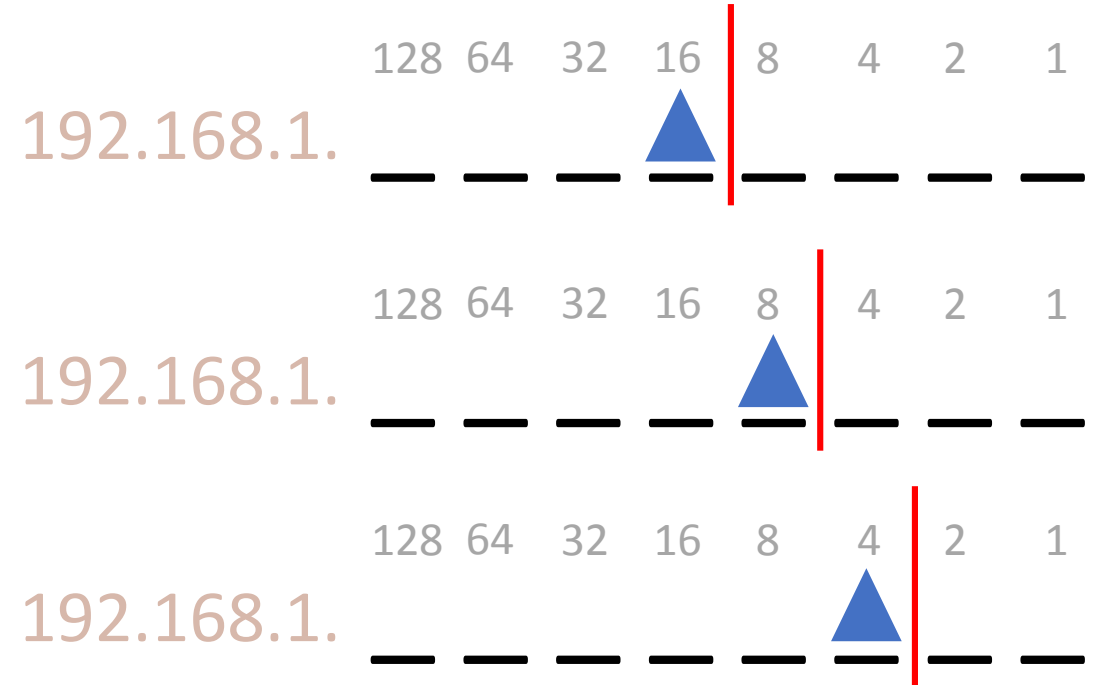
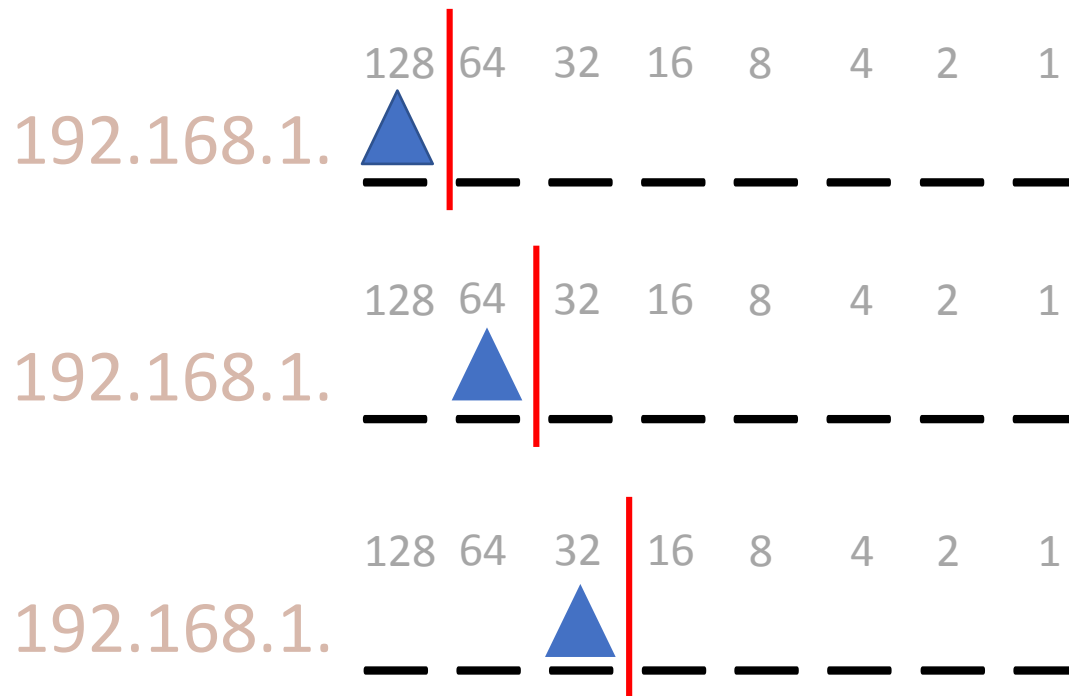


2. Determine # of Bits to Move the Mask



3. Move the Mask and Identify the Delta

The delta is the digit to the immediate left of the subnet mask dividing line



Let's Practice Moving the Mask

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Subnets
you need

Bits to move
the mask

255.255.255.0 /24

192.168.1.

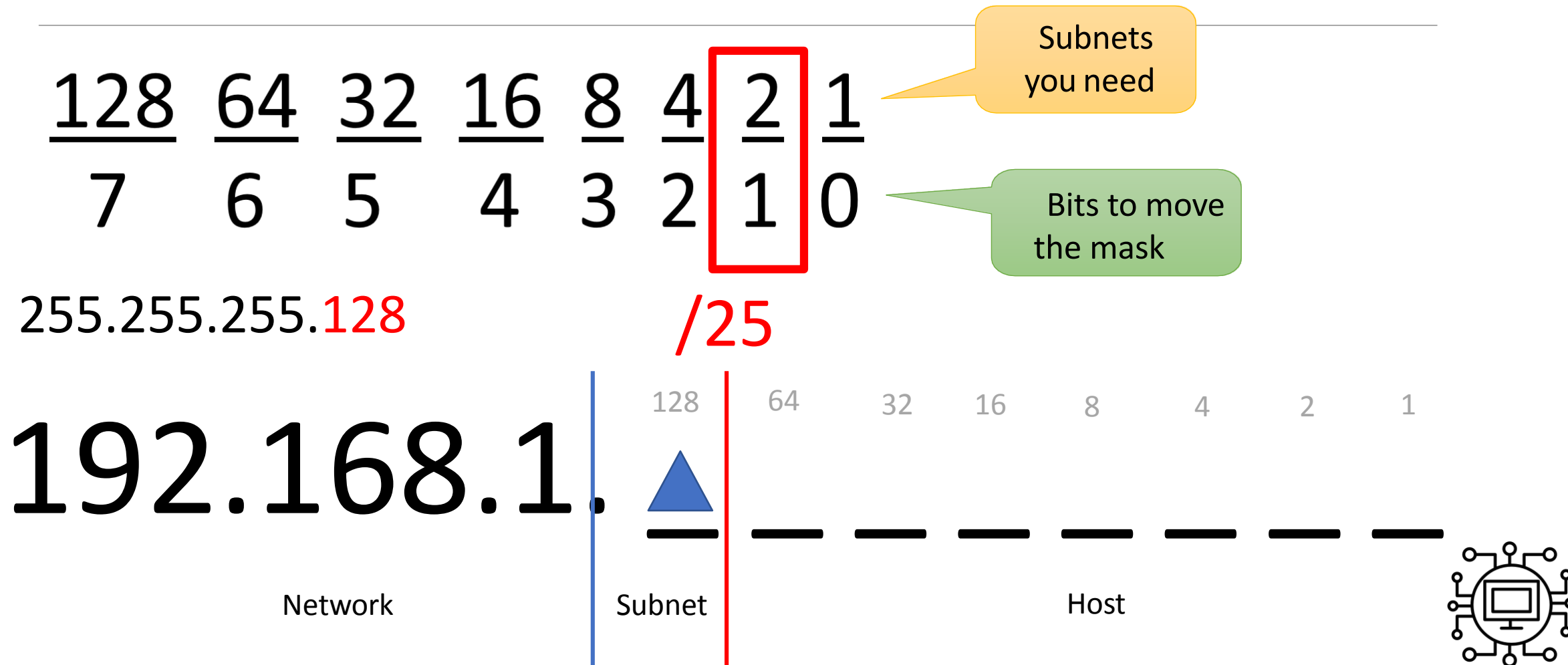
Network

128 64 32 16 8 4 2 1

Host



Let's Practice Moving the Mask (cont'd)



Let's Practice Moving the Mask (cont'd)

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Subnets
you need

Bits to move
the mask

255.255.255.192

/26

192.168.1.

Network



Subnet

Host



Let's Practice Moving the Mask (cont'd)

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Subnets
you need

Bits to move
the mask

255.255.255.224

/27

192.168.1.

Network

Subnet

Host



Let's Practice Moving the Mask (cont'd)

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Subnets
you need

Bits to move
the mask

255.255.255.240

/28

192.168.1.

Network



Subnet

Host



Let's Practice Moving the Mask (cont'd)

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Subnets
you need

Bits to move
the mask

255.255.255.248

192.168.1.

Network

128 64 32 16 8 4 2 1

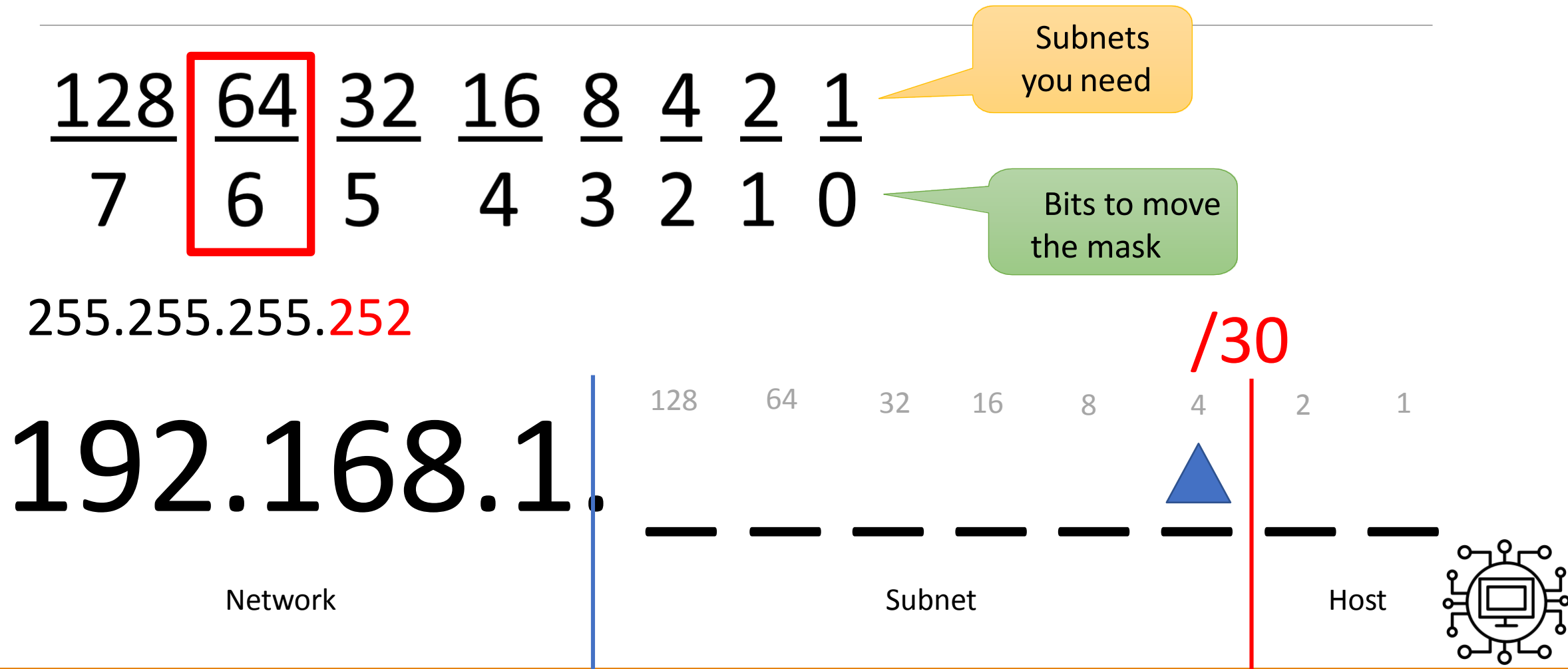
/29

Subnet

Host



Let's Practice Moving the Mask (cont'd)

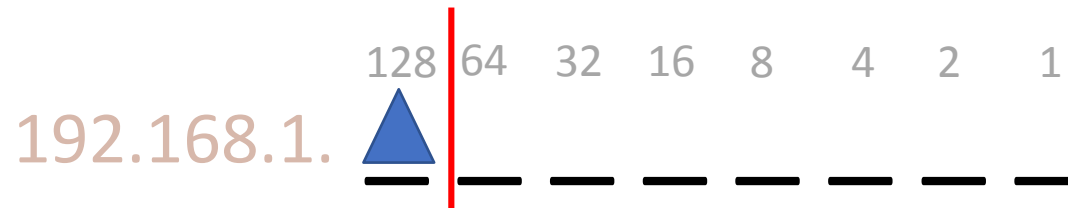


4. Use The Delta to Calculate Subnet IDs and IP Ranges

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



Delta is 128

Subnet 0 = 192.168.1.0

Subnet 128 = 192.168.1.128

Next subnet = 192.168.1.256

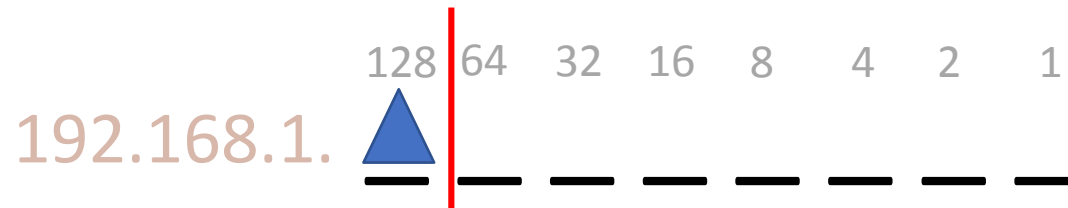


4. Use The Delta to Calculate Subnet IDs and IP Ranges

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



Delta is 128

Subnet 0 = 192.168.1.0

Subnet 128 = 192.168.1.128

~~Next subnet = 192.168.1.256~~

Out of Range!

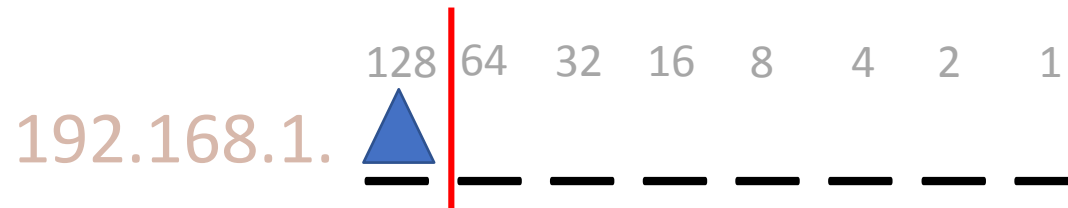


4. Use The Delta to Calculate Subnet IDs and IP Ranges

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



Delta is **128**

Subnet **0** = 192.168.1.**0**

1st IP = 192.168.1.1

Last IP = 192.168.1.126

Broadcast = 192.168.1.127

Subnet **128** = 192.168.1.**128**

1st IP = 192.168.1.129

Last IP = 192.168.1.254

Broadcast = 192.168.1.255



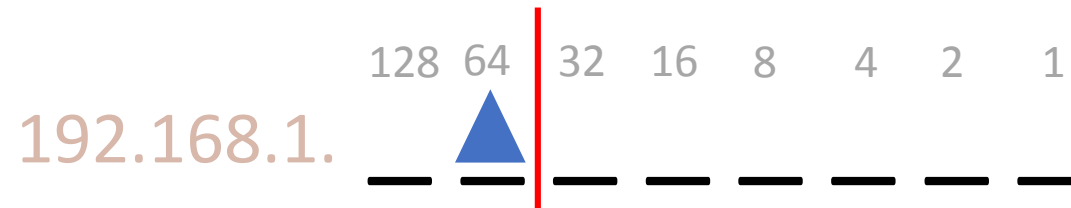
4. Let's Try Another Delta

Delta is 64

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



4. Let's Try Another Delta

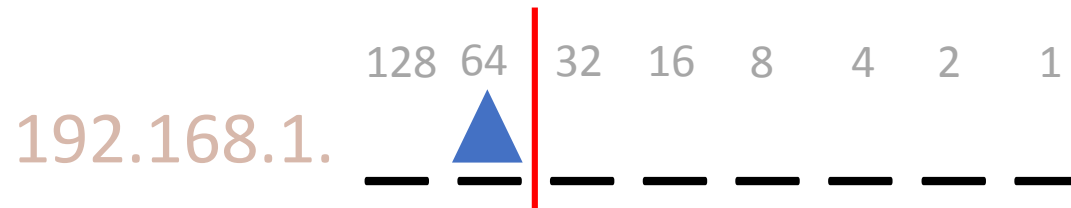
Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range

Delta is 64

Subnet 0	= 192.168.1.0
1 st IP	= 192.168.1.1
Last IP	= 192.168.1.62
Broadcast	= 192.168.1.63

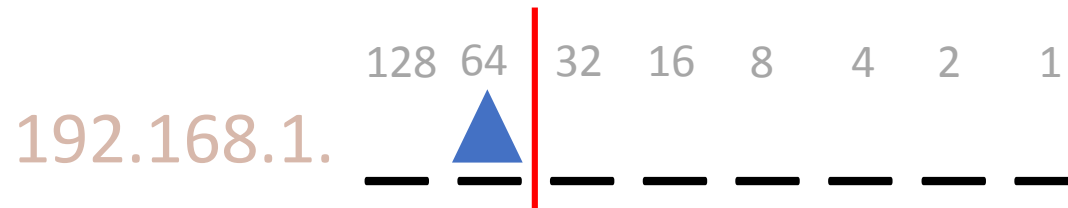


4. Let's Try Another Delta

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



Delta is 64

Subnet 0	= 192.168.1.0
1 st IP	= 192.168.1.1
Last IP	= 192.168.1.62
Broadcast	= 192.168.1.63

Subnet 64	= 192.168.1.64
1 st IP	= 192.168.1.65
Last IP	= 192.168.1.126
Broadcast	= 192.168.1.127

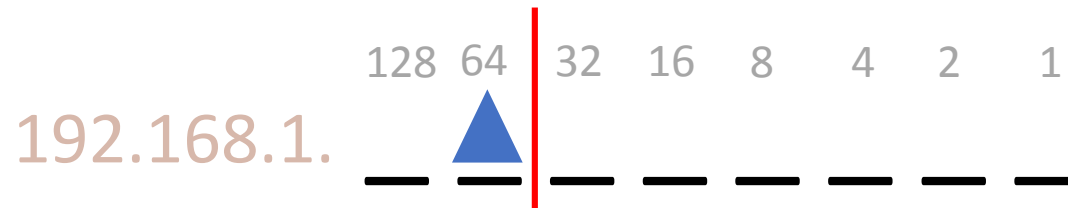


4. Let's Try Another Delta

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



Delta is 64

Subnet 0	= 192.168.1.0
1 st IP	= 192.168.1.1
Last IP	= 192.168.1.62
Broadcast	= 192.168.1.63

Subnet 64	= 192.168.1.64
1 st IP	= 192.168.1.65
Last IP	= 192.168.1.126
Broadcast	= 192.168.1.127

Subnet 128	= 192.168.1.128
1 st IP	= 192.168.1.129
Last IP	= 192.168.1.194
Broadcast	= 192.168.1.195

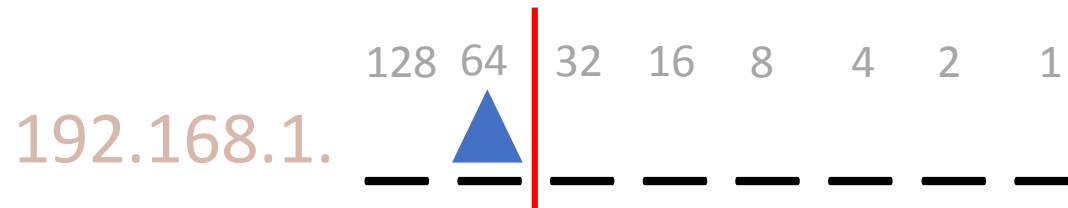


4. Let's Try Another Delta

Start at 0 (subnet 0)

Then increment subnet IDs by the delta

Stop when you are out of range



Delta is 64

Subnet 0	= 192.168.1.0
1 st IP	= 192.168.1.1
Last IP	= 192.168.1.62
Broadcast	= 192.168.1.63

Subnet 64	= 192.168.1.64
1 st IP	= 192.168.1.65
Last IP	= 192.168.1.126
Broadcast	= 192.168.1.127

Subnet 128	= 192.168.1.128
1 st IP	= 192.168.1.129
Last IP	= 192.168.1.194
Broadcast	= 192.168.1.195

Subnet 196	= 192.168.1.196
1 st IP	= 192.168.1.197
Last IP	= 192.168.1.254
Broadcast	= 192.168.1.255



The Delta in Action



Delta 128 Example

New subnet mask = /25

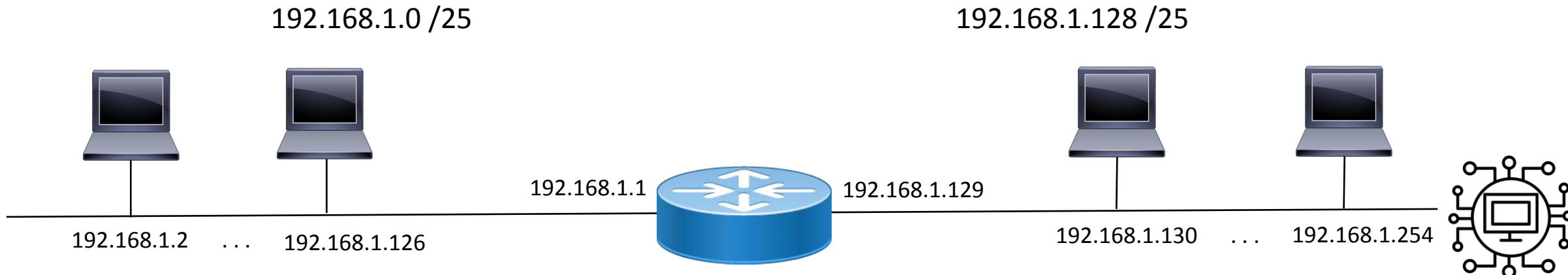
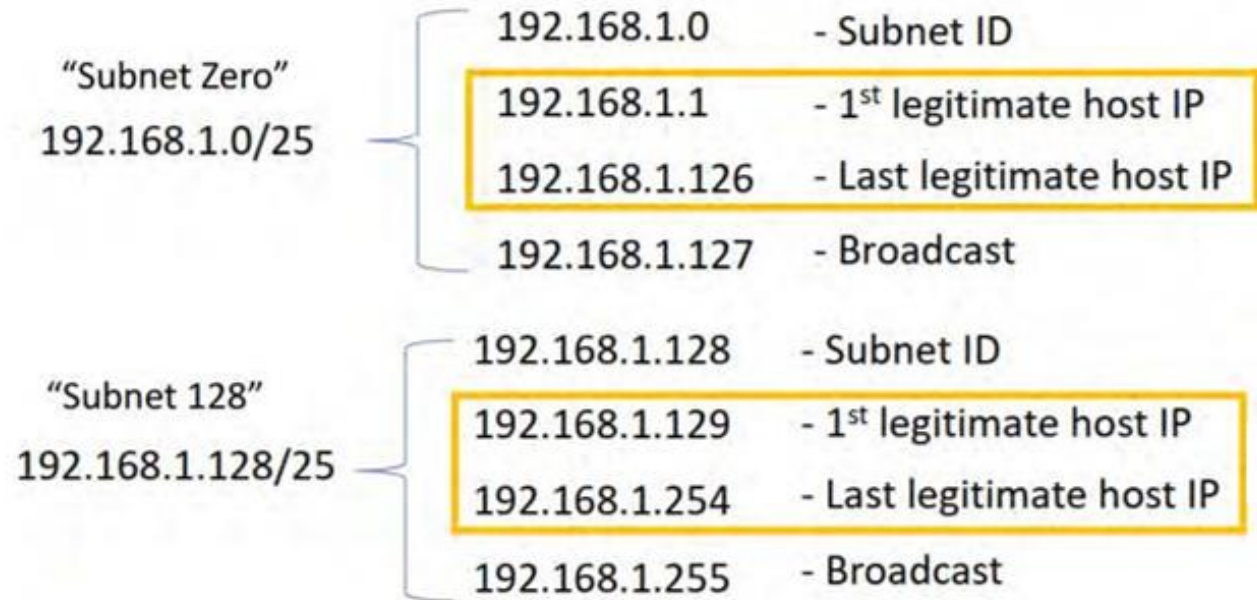
255.255.255.128

Delta = 128

Subnet IDs increment by 128

Create a DHCP scope based on the range of subnet IPs

- Give the router the 1st or last legitimate host IP
- Exclude a few addresses for servers, etc.
- Let the rest of the addresses be part of the active DHCP pool



Things to Notice About the Numbers

Regardless of the Delta:

The **Subnet ID** is ALWAYS an **EVEN** number

The **1st** legitimate host IP is ALWAYS an **ODD** number

The **last** legitimate host IP is ALWAYS an **EVEN** number

The **broadcast** is ALWAYS an **ODD** number

<https://www.site24x7.com/tools/ipv4-subnetcalculator.html>

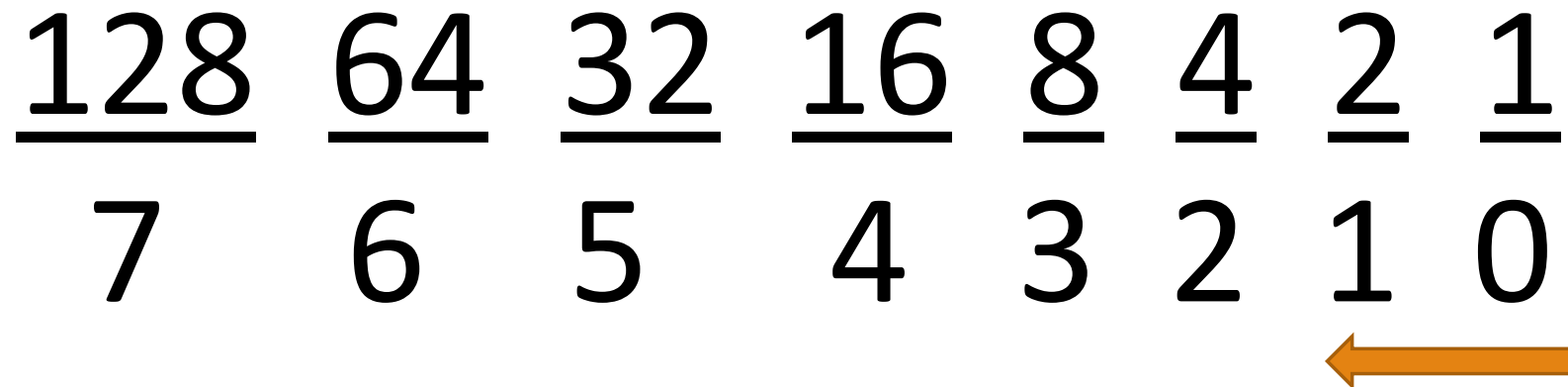


Subnetting Based on Hosts



Subnet Calculator Can Be Used for Hosts Too!

- When counting host bits, start from the very far right, past the last bit, and count to the left
- Always give hosts 2 extra addresses for the subnet ID and broadcast address
- Provide enough extra host IPs for growth
- Be careful that the bits for hosts does not overlap the needed bits for subnets



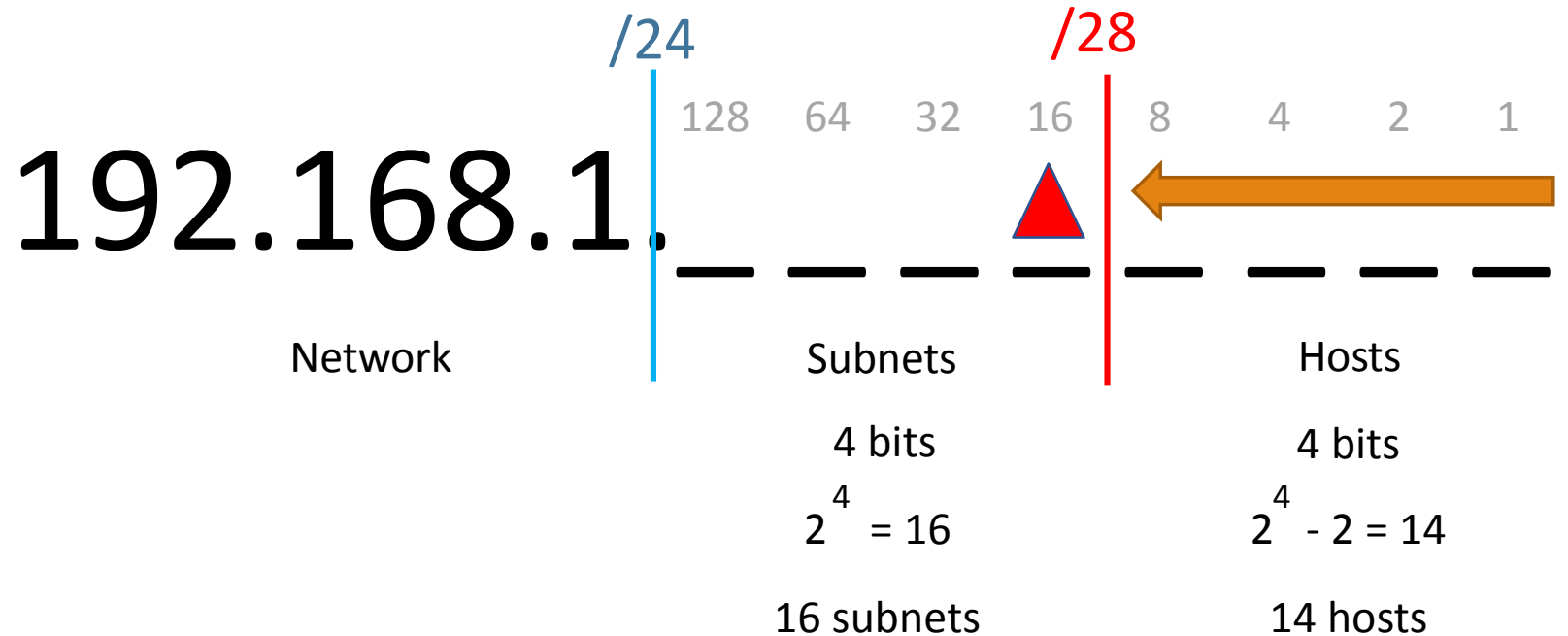
How Many Subnets Can You Have If Each Subnet Must Contain 12 Hosts?

Original Network = 192.168.1.0 /24

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many hosts + 2 you need per subnet

How many bits you will need for hosts



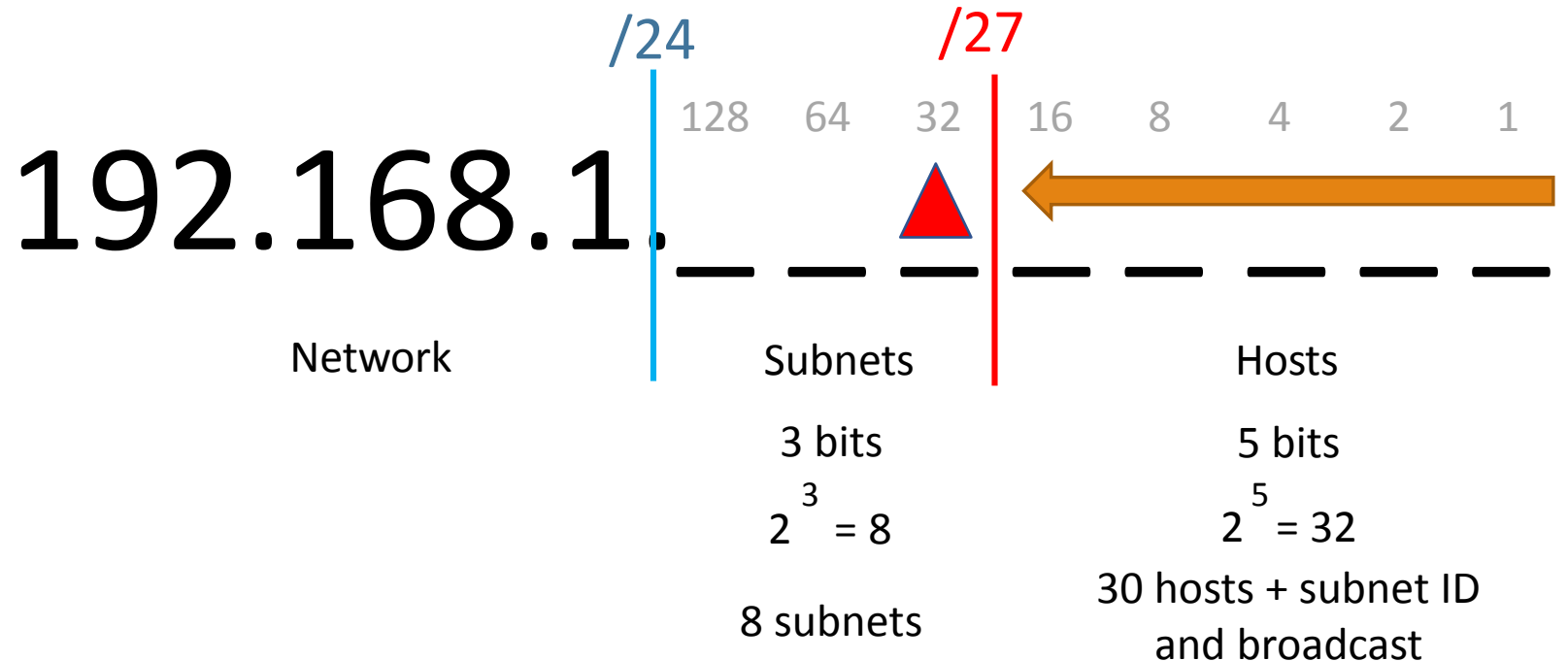
How Many Subnets Can You Have If Each Subnet Must Have 25 Hosts?

Original Network = 192.168.1.0 /24

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many hosts
+ 2 you need
per subnet

How many bits
you will need
for hosts



How Many Point-to-Point WAN Links Can You Provide Addresses For?

Original Network = 192.168.1.0 /24

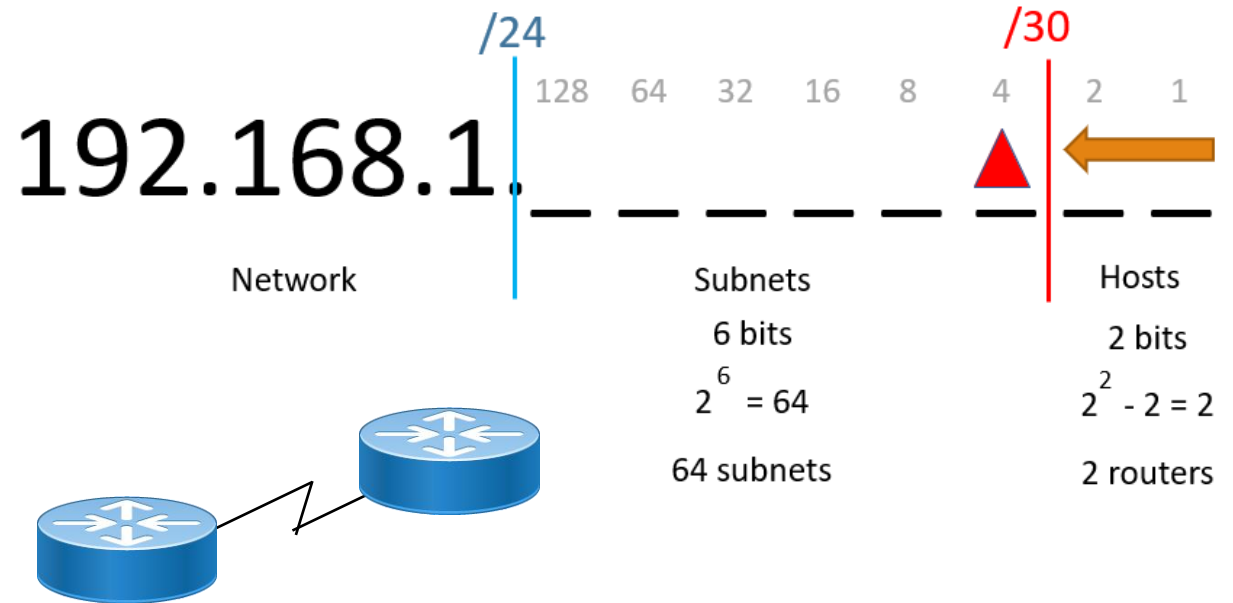
You only need 2 hosts (routers) per link (subnet)

Delta = 4

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many hosts + 2 you need per subnet

How many bits you will need for hosts



192	168	1	0	Subnet
			1	1 st
			2	Last
			3	B'cast
192	168	1	4	Subnet
			5	1 st
			6	Last
			7	B'cast



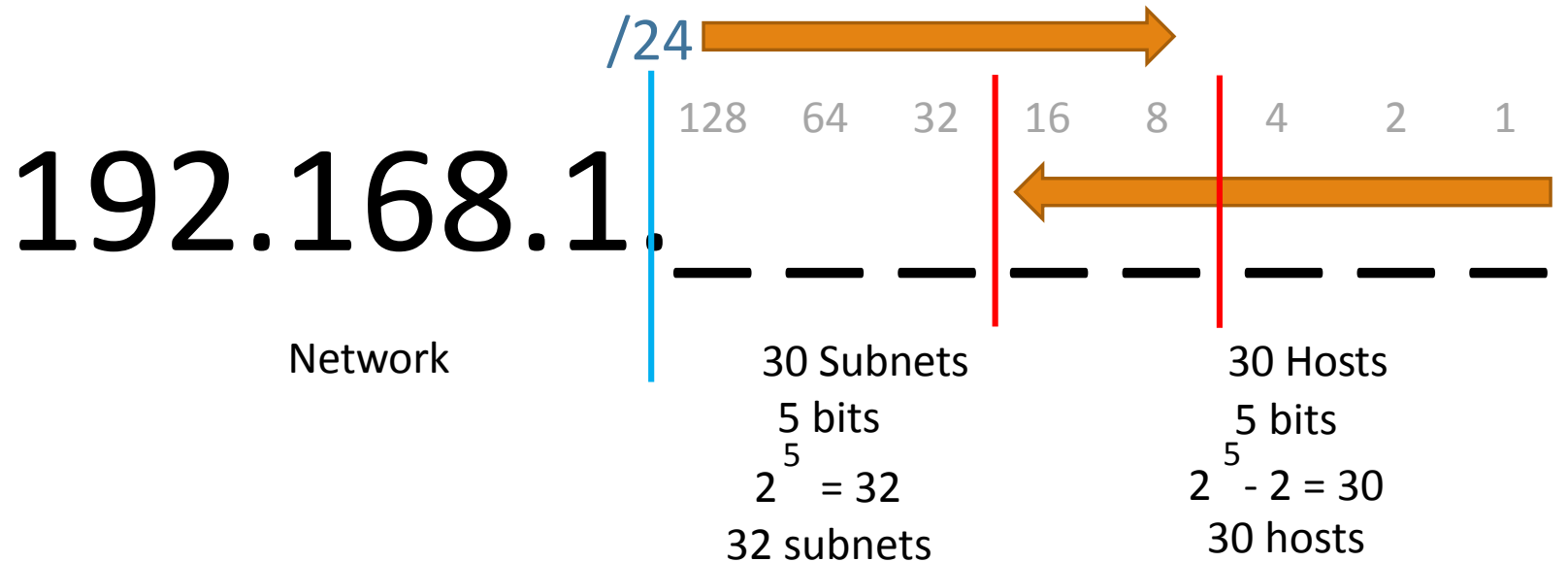
What If I Need 30 Subnets with 30 Hosts Each?

Original Network = 192.168.1.0 /24

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many hosts
+ 2 you need
per subnet

How many bits
you will need
for hosts



The math doesn't work! There's overlap
10 bits total is needed – that's too much!

Adjust your requirements or change your starting network / subnet mask

4.6 Subnetting in Other Octets

CCNA 200-301 v1.1

Module 4

Subnetting in the 3rd Octet

Subnetting Across Octets

Subnetting in the 2nd Octet

Subnetting in the 3rd Octet

Original subnet mask = /16

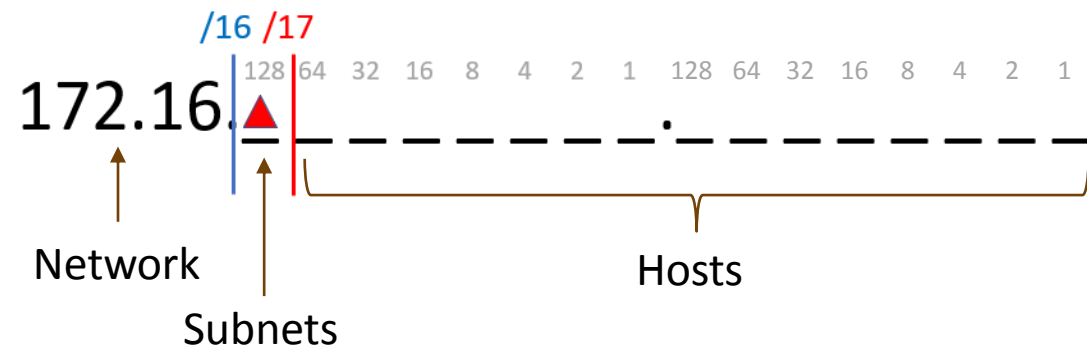
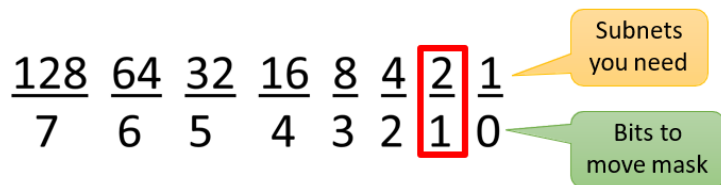
Need 2 subnets

Move mask 1 bit in third octet

Delta is 128 in third octet

New mask is /17 255.255.128.0

Subnet IDs increment by 128 in the 3rd octet



Each subnet will have 15 bits worth of hosts
 $2^{15} - 2 = 32,766$



Subnetting in the 3rd Octet

Original subnet mask = /16

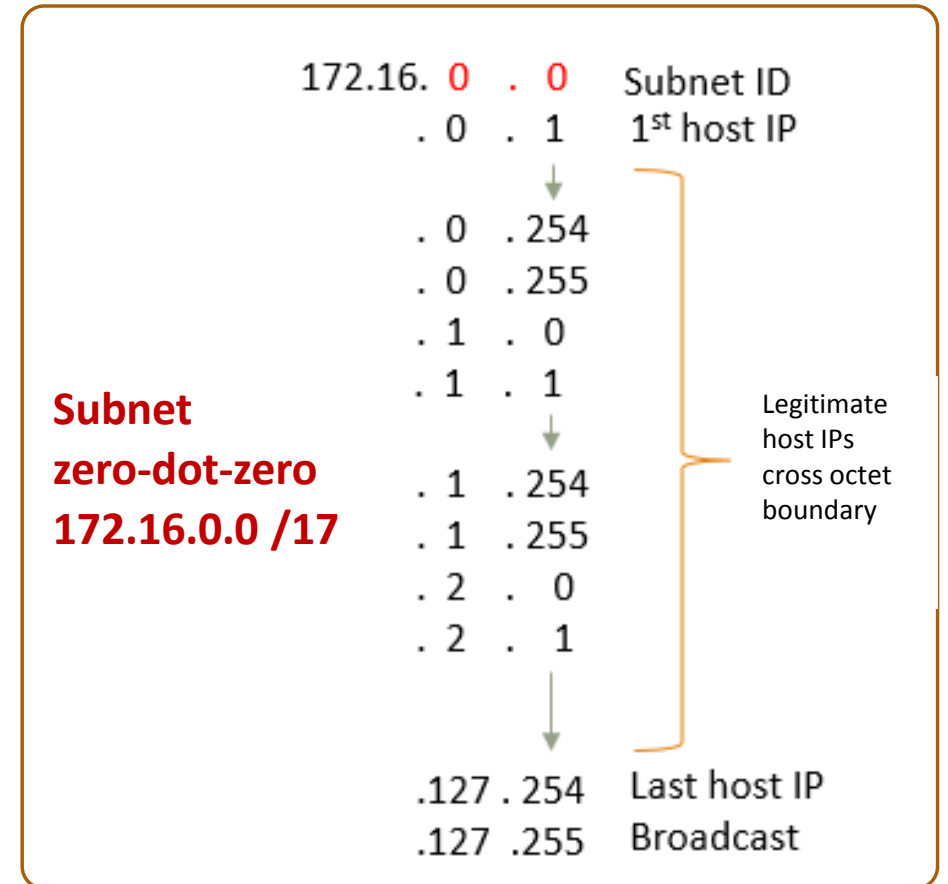
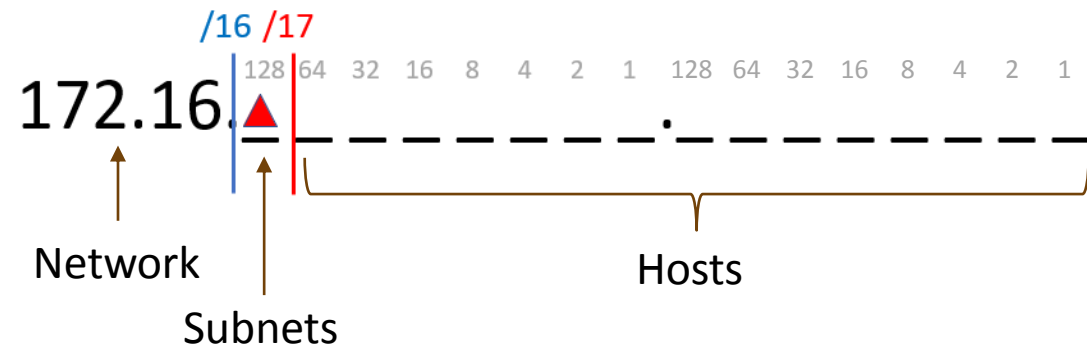
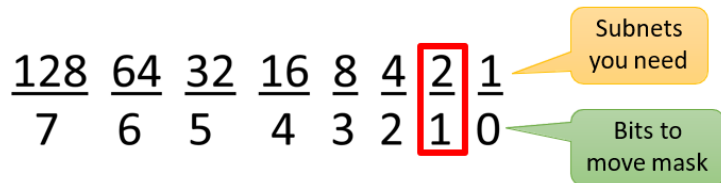
Need 2 subnets

Move mask 1 bit in third octet

Delta is 128 in third octet

New mask is /17 255.255.128.0

Subnet IDs increment by 128 in the 3rd octet



Subnetting in the 3rd Octet

Original subnet mask = /16

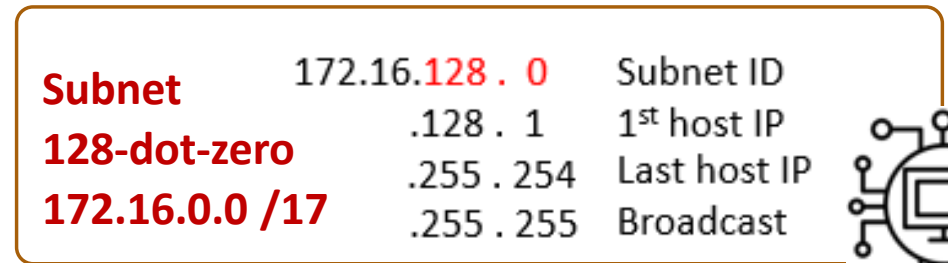
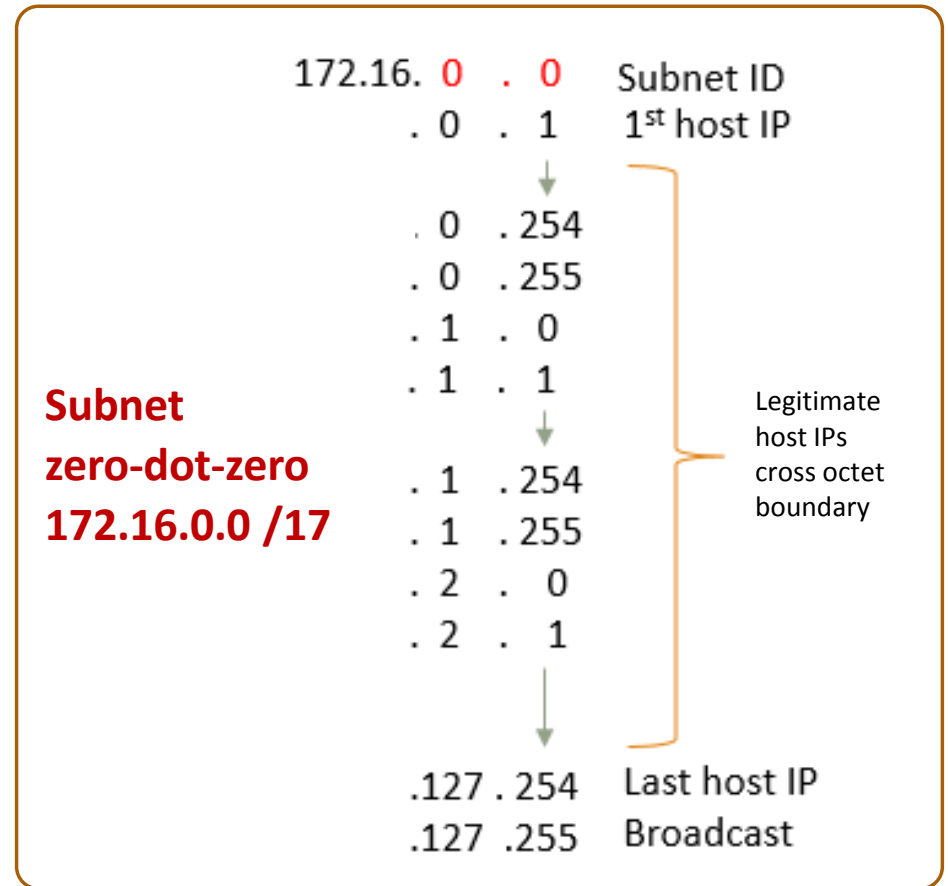
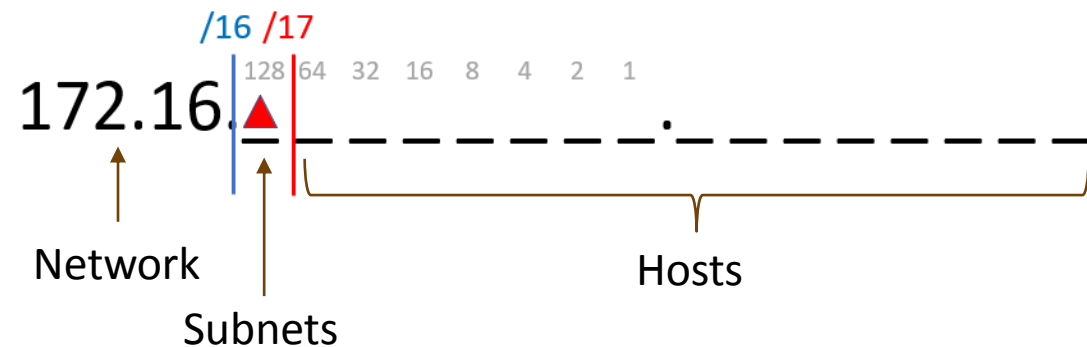
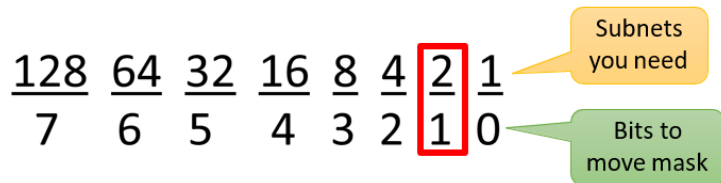
Need 2 subnets

Move mask 1 bit in third octet

Delta is 128 in third octet

New mask is /17 255.255.128.0

Subnet IDs increment by 128 in the 3rd octet



Subnetting Across TWO Octets

32768	16384	8192	4196	2048	1024	512	256		128	64	32	16	8	4	2	1
~32000	~16000	~8000	~4000	~2000	~1000	~500	~250									
15	14	13	12	11	10	9	8		7	6	5	4	3	2	1	0

How many you need

of Bits to move the mask

Two octet subnet calculator



Original subnet mask = /16 255.255.0.0

"I need 30 subnets"	- move the mask down 5 bits -	/16 → /21	255.255.248.0
"I need 100 subnets"	- move the mask down 7 bits -	/16 → /23	255.255.254.0
"I need about 250 subnets"	- move the mask down 8 bits -	/16 → /24	255.255.255.0
"I need about 500 subnets"	- move the mask down 9 bits -	/16 → /25	255.255.255.128
"I need about 600 subnets"	- move the mask down 10 bits	/16 → /26	255.255.255.192
"I FOR SURE need 1025 subnets"	- move the mask down 11 bits	/16 → /27	255.255.255.224

Always leave room for growth!

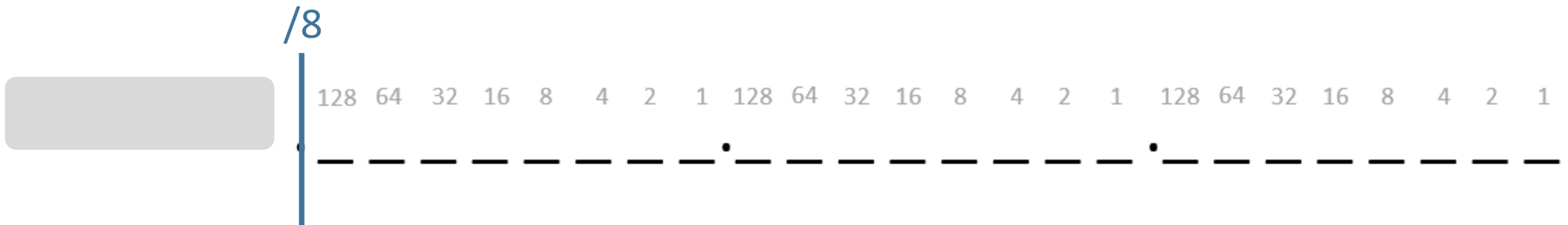


Subnetting in the 2nd Octet

Start with a /8 default subnet mask

Use the subnetting calculator (one or two octet) as you normally would

But move the mask from the /8 position down to the right as needed



Subnetting in the 2nd Octet Example

Original subnet mask = /8

Need 2 subnets

Move mask 1 bit in second octet

Delta is 128 in second octet

New mask is /9 (255.128.0.0)

Subnet IDs increment by 128 in the 2nd octet

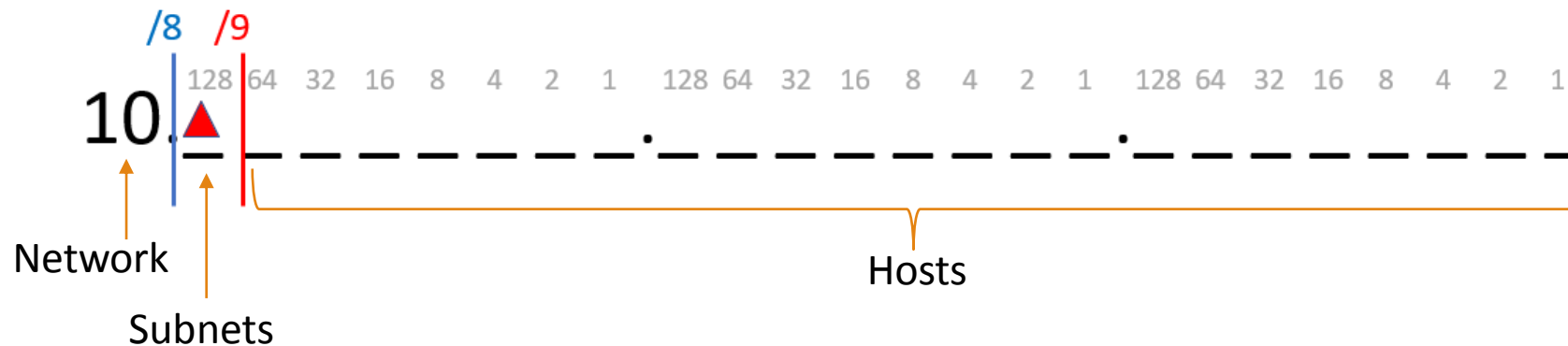
128	64	32	16	8	4	2	1
7	6	5	4	3	2	1	0

Subnets
you need

Bits to
move mask

10. 0 . 0 . 0 Subnet ID
 . 0 . 0 . 1 1st host IP
 .127.255.254 Last host IP
 .127.255.255 Broadcast

10.128. 0 . 0 Subnet ID
 .128. 0 . 1 1st host IP
 .255.255.254 Last host IP
 .255.255.255 Broadcast



Subnetting Across Octets 2 and 3 Example

Original subnet mask = /8

Need 4000 subnets

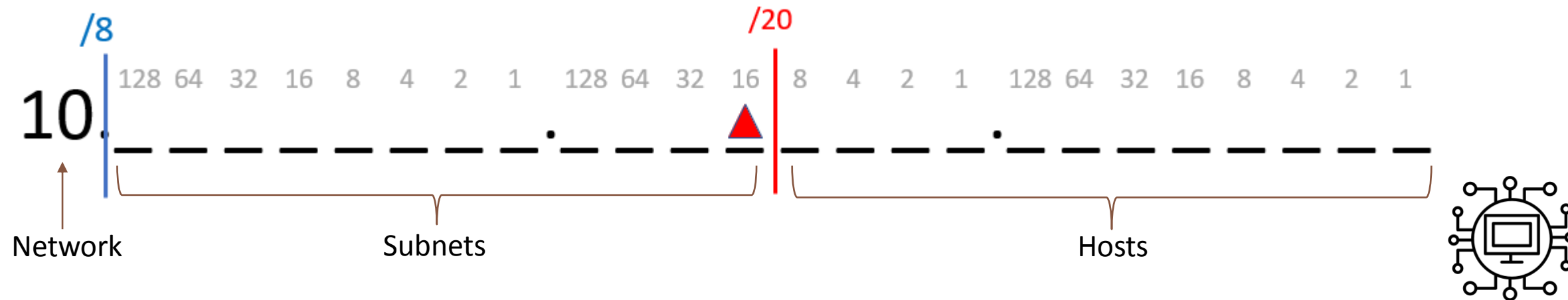
Move mask 12 bits down into the third octet

New mask is /20 (255.255.240.0)

Delta is 16 in third octet

Subnet IDs increment by 16

Each subnet has 12 bits for hosts = $2^{12} - 2 = 4094$



Subnetting Across Octets 2 and 3

Original subnet mask = /8

Need 4000 subnets

Move mask 12 bits down into the third octet

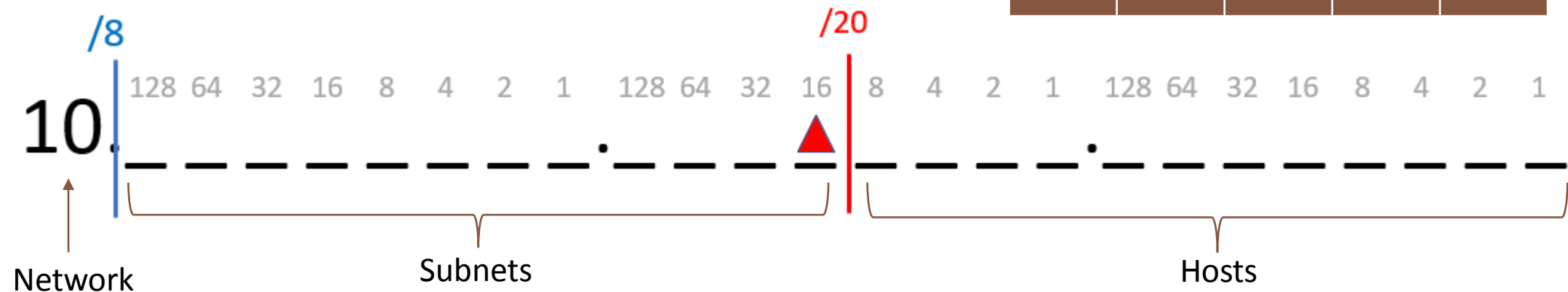
New mask is /20 (255.255.240.0)

Delta is 16 in third octet

Subnet IDs increment by 16

Each subnet has 12 bits for hosts = $2^{12} - 2 = 4094$

10	0	0	0	Subnet
	0	0	1	1st
	0	15	254	Last
	0	15	255	B'cast
10	0	16	0	Subnet
	0	16	1	1 st
	0	31	254	Last
	0	31	255	B'cast
10	0	32	0	Subnet



We don't subnet in the first octet!

4.7 Supernetting

CCNA 200-301 v1.1

Module 4

Supernetting

Supernet the Supernets

Supernetting

The opposite of subnetting

Move the mask to the **left** instead of the right

- Instead of dividing by 2, you multiply by 2

Combine multiple networks together to make a larger network

The mask gets shorter, not longer

The number of networks becomes fewer

The number of hosts per network becomes more

We use supernetting to:

- Create bigger subnets
- Aggregate routes to make router tables smaller



Supernet Example

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Number of
networks to
combine

Number of bits to
reduce the mask

We wish to combine 2 networks into 1 supernet

The original subnet mask will be reduced by 1

The new supernet ID will be the same as the lowest original network

But the subnet mask will be shorter

Original Networks					Supernet 192.168.0.0 /23
192	168	0	0	/24	
192	168	1	0	/24	



Supernetting the Supernets

AKA Route Aggregation

You can combine supernets to create even bigger supernets!

Or, you can use the subnet calculator to directly create larger supernets

Supernet IDs, like subnet IDs, are ALWAYS an even number

Original Networks				
192	168	0	0	/24
192	168	1	0	/24
192	168	2	0	/24
192	168	3	0	/24
192	168	4	0	/24
192	168	5	0	/24
192	168	6	0	/24
192	168	7	0	/24

Supernet
192.168.0.0 /23

Supernet
192.168.2.0 /23

Supernet
192.168.4.0 /23

Supernet
192.168.6.0 /23

Supernet
192.168.0.0 /22

Supernet
192.168.4.0 /22

Supernet
192.168.0.0 /21

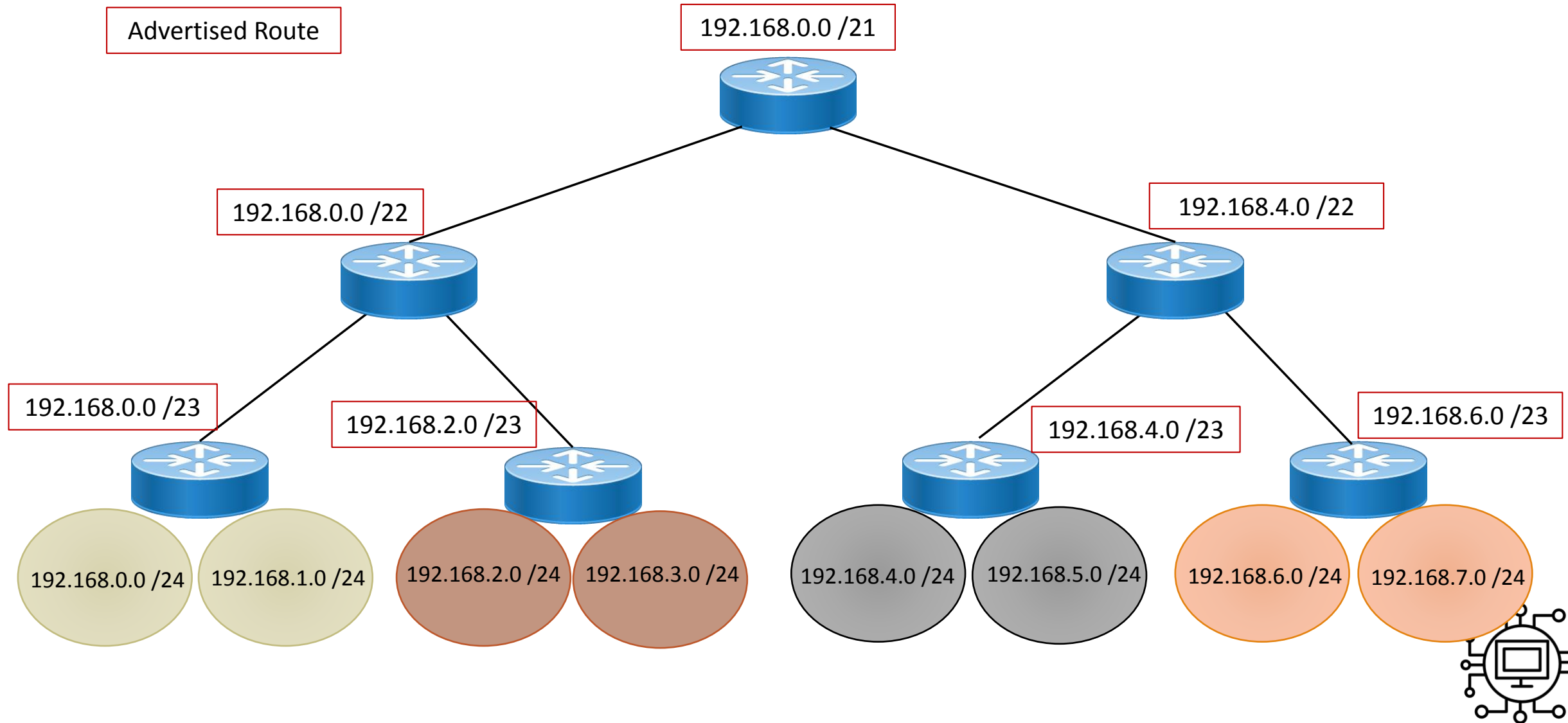
128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

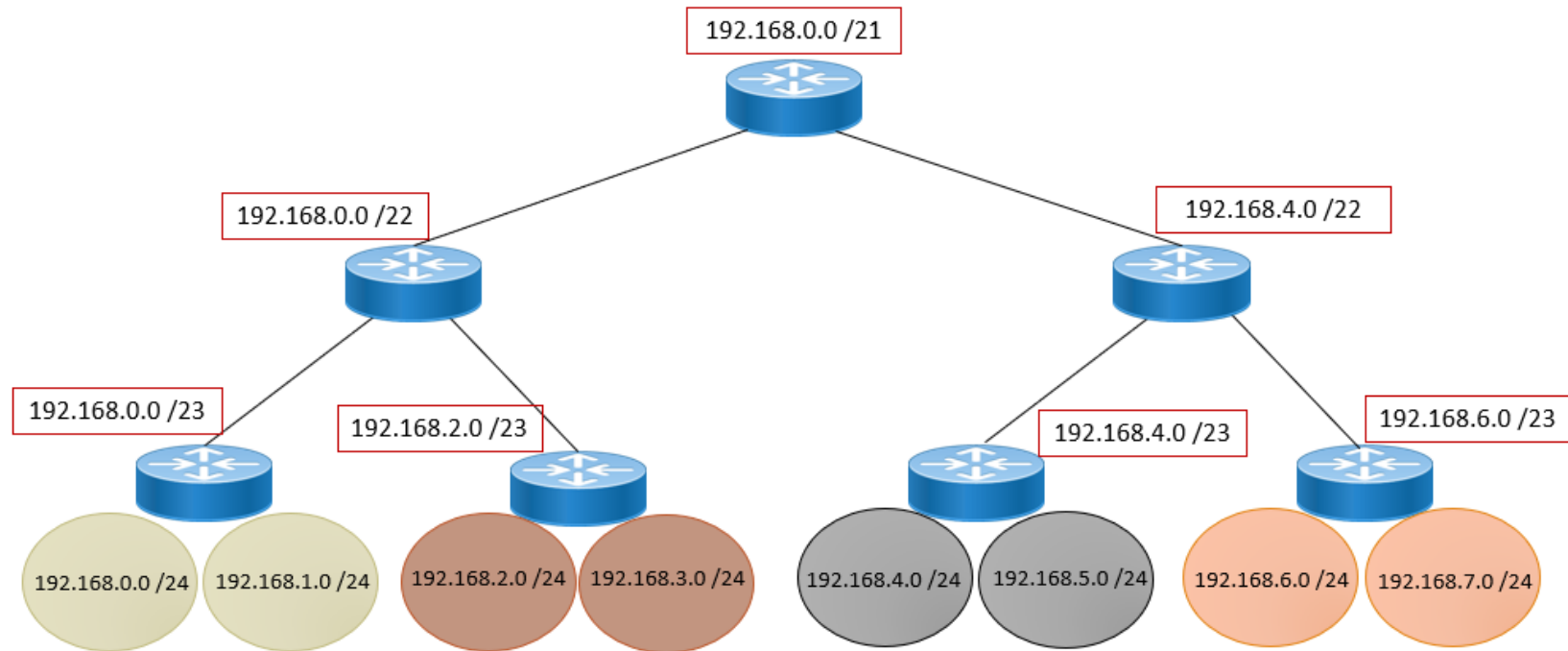
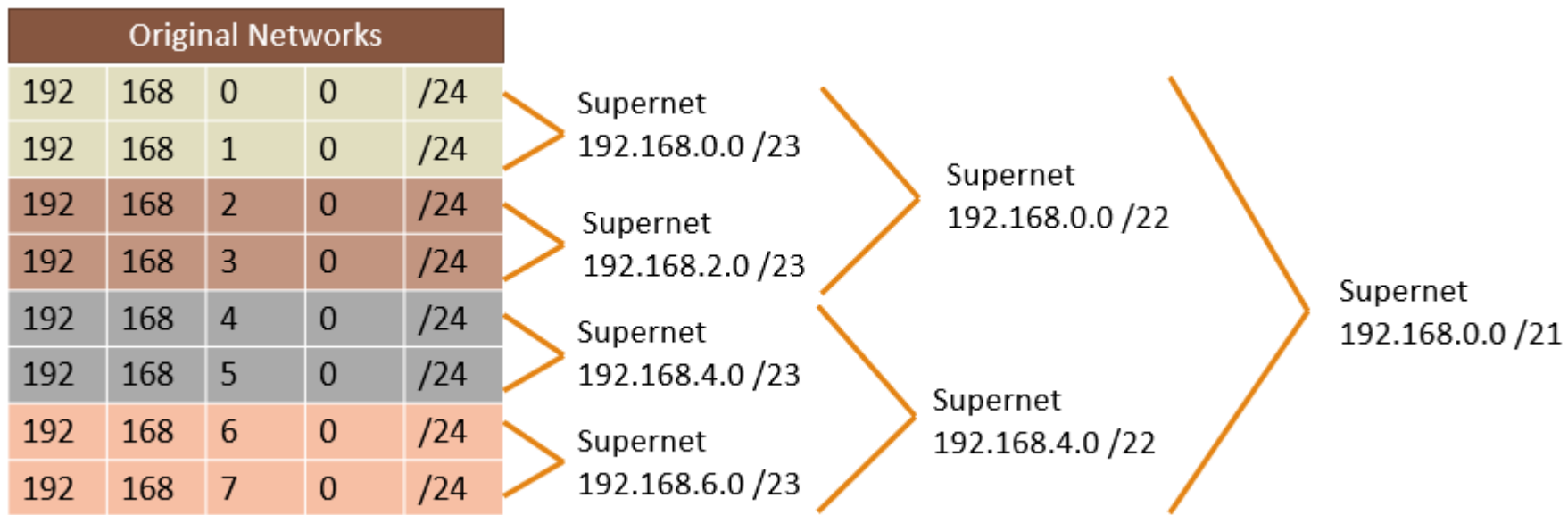
Number of
subnets you wish
to combine

Number of bits to
reduce the mask



Supernets in Action





4.8 IPv6

CCNA 200-301 v1.1

Module 4

IPv6 Addressing

IPv6 Subnet Mask and Subnets

IPv6 Addressing Mechanisms

Neighbor Discovery Protocol (NDP)

IPv6 Unicast Address Types

IPv6 Other Address Types

Compressing IPv6 Addresses

IPv6 Host ID

Modified EUI-64

IPv4 – IPv6 Transition Mechanisms

IPv6 Addressing

Created to resolve the eventual depletion of IPv4 public addresses

128 bits in length

- Written in hexadecimal as eight hextets separated by colons (:)
- Each hextet = 4 hex numbers
- 1 hex number = 4 binary bits

Example:

- 2601:140a:8780:43f0:2d1e:9ebb:92f0:d6e9
- hextet

NOTE: Although the subnet mask (prefix) can be variable, at the host level it is always /64

Hexadecimal numbers can be written in upper or lower case



IPv6 Subnet Mask and Subnets

Always written in CIDR

Hosts always use /64

- Originally could be any bit length

If the ISP permits, the 4th hextet can be used by the enterprise customer to create internal subnets

ISP are often called “48s” because they use the first 48 bits of the IP address



IPv6 Addressing Mechanisms

Mechanism	Description
Statically assigned	Configured by an administrator
Self-assigned	Link Local address automatically assigned by operating system
DHCPv6	DHCP, but with IPv6 addresses
Stateless address autoconfiguration (SLAAC)	1. Node transmits a multicast Router Solicitation (RS) to find its default gateway 2. Router responds with a Router Advertisement (RA) that contains the subnet ID 3. Node appends its host ID to complete the address Subnet mask is always /64



Neighbor Discovery Protocol (NDP)

Layer 2 protocol in IPv6

Used by a node to discover:

- The subnet it's on
- Neighbors

NDP Function	Description
Router Solicitation (RS)	Sent by nodes when they boot up to find routers in a local segment Sent to the All Router IPv6 multicast address (FF02::2)
Router Advertisement (RA)	Sent periodically or in response to RS Router identifies itself and the subnet ID Sent to the all-nodes link-local multicast group address (FF02::1)
Neighbor Solicitation (NS)	Discover neighbor's link layer address (similar to IPv4 ARP)
Neighbor Advertisement (NA)	Response to NS
Redirect	Router refers local nodes to a different default gateway



IPv6 Unicast Address Types

These unicast types are for a single interface

Global Unicast

- Publicly routable
- Comparable to IPv4 public IP addresses
- Assigned by IANA and used on public networks
- Have a prefix of 2000::/3
 - All addresses begin with binary 001 (first hexet range is 2000 – 3FFF)

Unique Local

- Used in private networks
- Routable on your intranet, but not on the Internet
- Comparable to IPv4 private addresses
- Have a prefix of FC00::/7
 - First hexet range is FC00 – FDFF
 - You will often see these written as FD00::/8



IPv6 Unicast Address Types (cont'd)

Link Local – similar to IPv4 APIPA (self assigned) addresses

- Also a unicast address
- Used for sending packets over the local subnet
- Comparable to IPv4 169.254.0.0/16
- Routers do not forward packets with this addresses to other subnets
- IPv6 requires a link-local address to be assigned to every network interface on which the IPv6 protocol is enabled
- Have a prefix of FE80::/10
- Example: fe80::10:18ff:fe12:2001%4

Interface index (number) on a computer

The interface index is used in both IPv4 and IPv6



IPv6 Other Address Types

Anycast

- Multiple servers or nodes share the same IP address
- Router routes packet to nearest healthy server
- Used in DNS and Content Delivery Networks (CDNs)
- ff00::/8

Multicast

- Address block FF00::/8 and FF00::/12 for forwarding packets to a group address
- FF02::1 (all nodes)
- FF02::2 (all routers)

Loopback

- Used when a host talks to itself over IPv6
- One program sends data to another program on the same host
- ::1/128

Unspecified

- Host has not yet learned/been assigned its own address
- ::/128



Compressing IPv6 Addresses

Done to simplify the address

You can omit *leading* zeros in a hextet

- You cannot omit trailing zeros
- E.g. 0001:0020:0340:000A:1234:abcd:0023:0001 → 1:20:340:A:1234:abcd:23:1
- fe80:0000:0000:0000:a299:9bff:fe18:50d1 → fe80:0:0:0:a299:9bff:fe18:50d1

:0: represents one hextet that is all 0s

- e.g. 1234:0000:5678:0000:abcd:efef:0000:cdcd can be expressed as 1234:0:5678:0:abcd:efef:0:cdcd

:: represents any number of contiguous 0s in an address

- 2001:EB8:C1:2200:1::331/64
- Can only have one :: in an address
- If you have already used :: in the address, you must represent subsequent 0000 as :0: or :0000:



Compressed IPv6 Address Examples

Using :: for Consecutive Zeros

- 2001:0db8:0000:0000:0000:0000:0000:0001 → 2001:db8::1
- fe80:0000:0000:0000:a299:9bff:fe18:50d1 → fe80::a299:9bff:fe18:50d1
- Fe80:0000:0000:0000:0000:0000:0000:0000 → fe80::

Omitting Leading Zeros

- 2001:0db8:0000:0000:0000:ff00:0042:8329 → 2001:db8::ff00:42:8329

Combining Both Methods

- 2001:0db8:0000:0000:0008:0800:200c:417a → 2001:db8::8:800:200c:417a



IPv6 Host ID



Obtaining an IPv6 Host ID

The host ID in IPv6 can be:

- Manually configured
- Self-assigned
- Randomly assigned by the operating system (Microsoft)
- Calculated from the interface MAC address

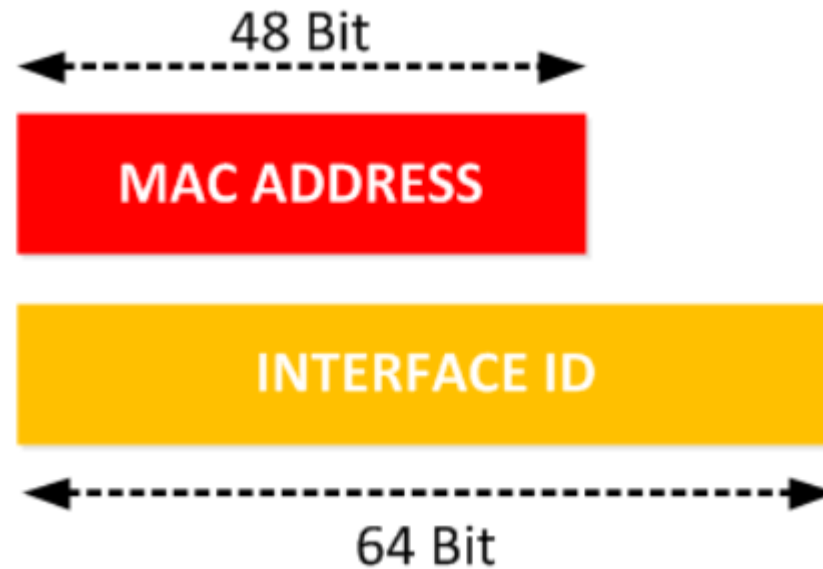


Using the MAC Address as the IPv6 Host ID

An IPv6 device can use its MAC address to generate a unique 64-bit interface (host) ID

However, a MAC address is only 48 bits but the interface ID must be 64 bits

We need to somehow add 16 bits to the MAC address to make it the right size!



EUI-64

AKA Modified EUI-64

A common way for a node to create its own host ID

The 48-bit MAC address is split into two 24-bit halves

- Example MAC address: AA:BB:CC:DD:EE:FF
- Split into: AA:BB:CC and DD:EE:FF

Insert the fixed 16-bit value FFFE between the two halves.

- Result: AA:BB:CC:FF:FE:DD:EE:FF



EUI-64 (cont'd)

Toggle the Universal/Local bit (7th bit of the first byte)

- If 0, change to 1
- If 1, change to 0
- AA in binary = 1010 1010 AA:BB:CC:FF:FE:DD:EE:FF
- Toggle bit 7 from 1 to 0
- The first byte is now 1010 1000 (A8)

Combine the parts to create the new MAC address: A8:BB:CC:FF:FE:DD:EE:FF

Add the prefix to create the entire IPv6 address: 2001:0db8:85:a3:A8BB:CCFF:FE:DD:EEFF /64



Microsoft IPv6 Host ID

Microsoft does not use EUI-64

- Instead, the OS randomly generates a host ID
- The ID is then used in SLAAC addressing or self-assignment

Microsoft also randomly creates temporary IPv6 addresses for security

- IPv6 SLAAC addresses are static and thus might become a security risk
- Temporary Global Unique addresses that change periodically are used instead

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix  . : hsd1.va.comcast.net
Description . . . . . : Broadcom 802.11n Network Adapter
Physical Address. . . . . : BC-85-56-F3-13-02
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2601:140:8780:43f0::4762(Preferred)
Lease Obtained. . . . . : Tuesday, October 26, 2021 12:57:15 PM
Lease Expires . . . . . : Friday, November 5, 2021 8:17:23 AM
IPv6 Address. . . . . : 2601:140:8780:43f0:2d1e:9ebb:92f0:d6e9(Preferred)
Temporary IPv6 Address. . . . . : 2601:140:8780:43f0:c9cc:e657:b99e:51a4(Preferred)
Link-local IPv6 Address . . . . . : fe80::2d1e:9ebb:92f0:d6e9%4(Preferred)
```



Configuring IPv6 on a Cisco Interface

Manually configure IPv6:

- `Switch(config-if)# ipv6 address 2001:db8:85a3::8a2e:370:7334/64`

Manually configure Modified EUI-64:

- `Router(config-if)# ipv6 address 2001:DB8:5:112::/64 eui-64`

Automatically configure Modified EUI-64:

- `Router(config-if)# ipv6 enable`
- `Router(config-if)# ipv6 address autoconfig`

Verify address:

- `Router# show ipv6 interface GigabitEthernet0/1`



IPv4 – IPv6
Transition
Mechanisms



Tunneling

A “tunnel” is a transmission in which one packet is hidden inside another packet

- Most frequently used in VPNs
- May or may not be encrypted

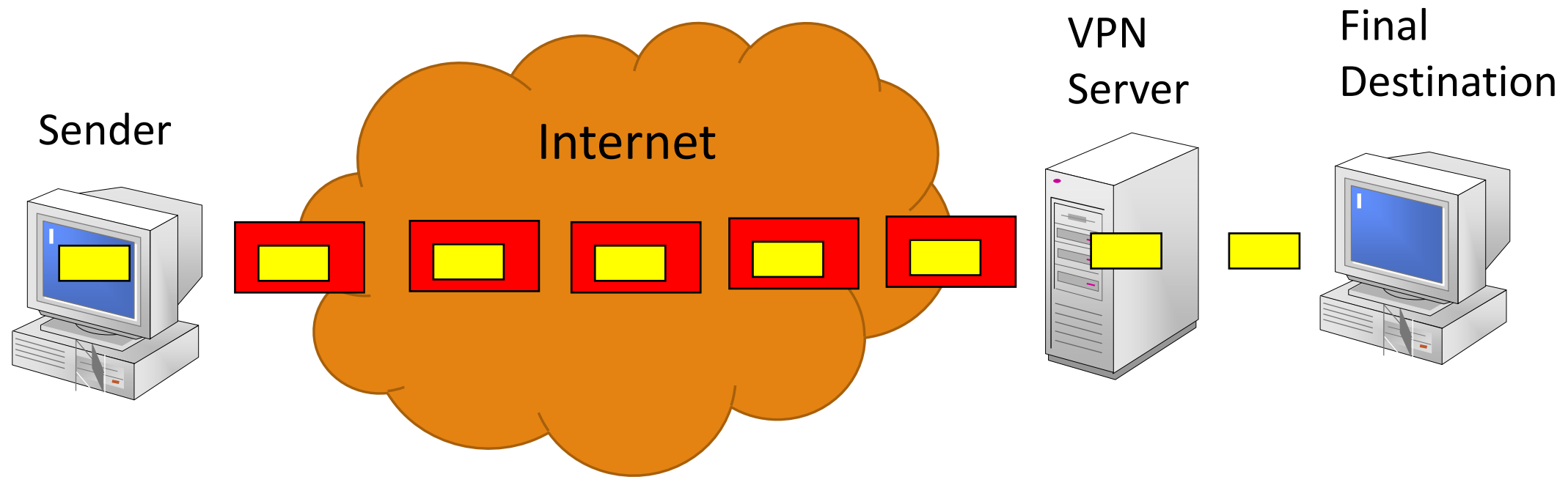
The same solution can be applied to tunnel IPv6 packets inside IPv4 packets

Different tunneling types:

- 6to4 - hide v6 packets inside v4 packets
- 6rd (6 rapid deployment) - a lightweight variant of 6to4
- ISATAP - dual stack nodes (typically Windows servers) act as 6-to-4 encapsulating proxies on a LAN
 - Windows clients automatically recognize the ISATAP server
 - Tunnel starts in the LAN
 - Tunnel is not encrypted



Tunnel Example



Dual Stack

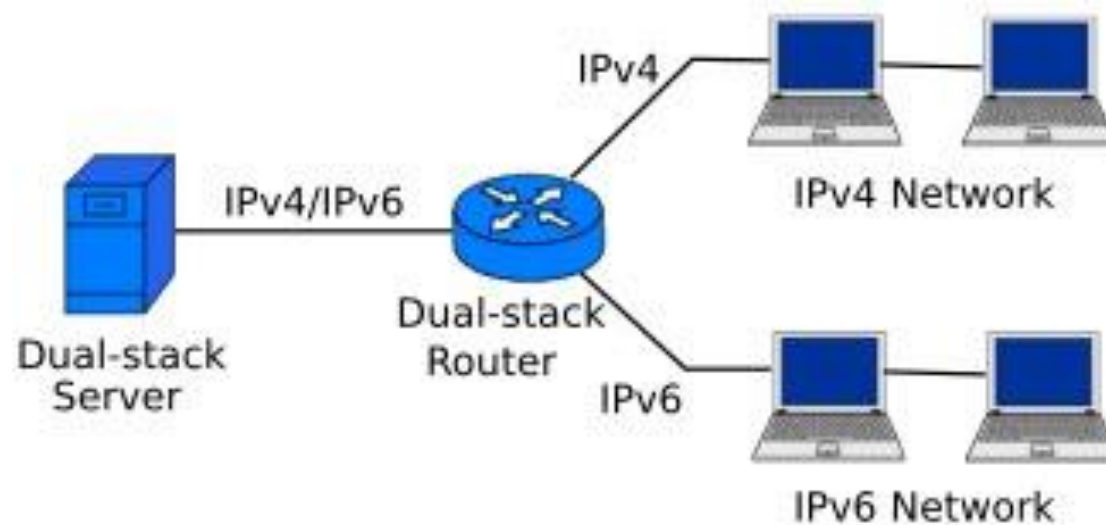
In dual-stack configuration, the server is configured for both IPv4 and IPv6

Can be implemented on a single interface or using multiple interfaces

- One for each protocol

End nodes and routers/switches run both protocols

- IPv6 is preferred



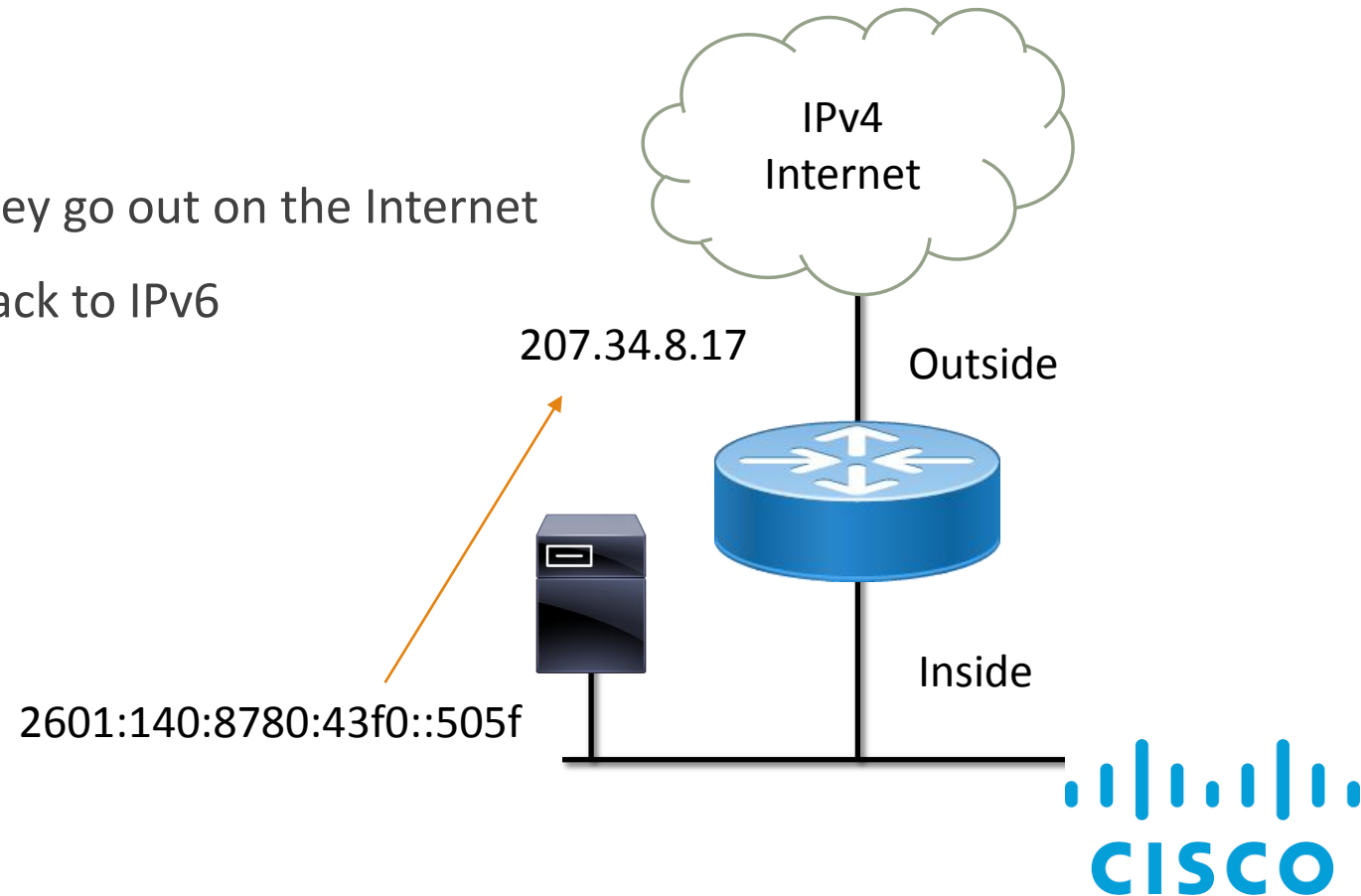
IPv6 – IPv4 NAT

IPv6 is on the “inside”

IPv4 is on the “outside”

IPv6 addresses are translated to IPv4 as they go out on the Internet

Returning IPv4 addresses are translated back to IPv6



Review

4.9 Review

Internet Protocol (IP)

CCNA 200-301 v1.1

Module 4

Review

An IP address is a layer 3 address that uniquely identifies a node on a network

IPv4 addresses are 32 bits, organized in 4 groups of 8 (octets)

192.168.1.100, 206.43.72.89, and 8.8.8.8 are examples of IPv4 addresses

IPv6 addresses are 128 bits long, organized in 12 groups of 4 hex numbers (hextets)

The subnet mask divides an IP address into network and host parts

A host uses its subnet mask to determine if a destination is on the same or different network

If the destination is on a different network, the host hands the packet to its default gateway

All hosts on the same network segment must use the same subnet mask to have the same point of reference



Review (cont'd)

Public IP addresses are globally unique and can be used on the Internet

Private IP addresses are:

Used to extend the lifespan of IPv4 global addresses

10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16

Only usable on a private network; they must be NAT or PAT translated for their traffic to go on the Internet

In Windows, you can use the command `ipconfig` to view IP address information

In Linux or macOS, you can use the command `ifconfig` to view IP address information



Review (cont'd)

A unicast is a one to one transmission

A multicast is one to a group

A broadcast is one to all

An anycast is a public IPv6 address that is shared by multiple nodes that offer the same service

An IPv6 link local address is a secondary IPv6 address specific to your local network segment

A loopback is a software interface used for testing, diagnostics, and to identify a router

The default IPv4 loopback address is 127.0.0.1

The default IPv6 loopback address is ::1



Review (cont'd)

A virtual IP address is one that is shared between two nodes

You can add logical subinterfaces on a router's FastEthernet physical interface

- This is called "Router on a stick"

Each subif has its own IP address and acts as the default gateway for a particular VLAN

In today's modern datacenters, router on a stick has been largely replaced with VLAN interfaces on Layer 3 switches



Review (cont'd)

Binary is a Base 2 numbering system

- When you move one digit to the right, you divide your number in half
- When you move one digit to the left, you double your number

The subnet mask is a series of contiguous binary 1s that end abruptly, followed by zeroes

- The dividing line between network ID and host ID is where the 1s end and the 0s begin

An IPv4 address and its subnet mask can be written binary to help calculate subnets and hosts



Review (cont'd)

The subnet calculator table is a convenient tool for:

- Customizing a subnet mask
- Calculating the number of bits to move a mask
- Calculating the number of available subnets and hosts
- Determining how to increment subnet IDs

Memorize this Tool!

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many
subnets you
need

How many bits
to move the
mask



Review (cont'd)

When you go down (to the right) a digit, you are subnetting your network into two

- Add a binary 1 to the subnet mask

When you go up (to the left) a digit, you are supernetting two networks together

- Remove a binary 1 from the subnet mask

You can extend the subnet calculator table to subnet across octets

Supernetting is the opposite of subnetting; instead of dividing a network by halves, you add networks together

Each time you combine two networks together to create a supernet, you make the subnet mask one bit shorter

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0



Review (cont'd)

Class A IPv4 addresses start with 0 for the high order (far left) bit

Class B IPv4 addresses start with 10 for the high order bits

Class C addresses start with 110 for the high order bits

Class D starts with 1110

Class E starts with 1111

The default subnet mask for a class A address is 255.0.0.0 /8

The default subnet mask for a class B address is 255.255.0.0 /16

The default subnet mask for a class C address is 255.255.255.0 /24

Class D and E addresses do not have a subnet mask



Review (cont'd)

Classless addressing uses a customized subnet mask

VLSM is classless addressing in action

CIDR expresses the subnet mask by how many 1s it has

Subnetting is the act of dividing a larger network into smaller subnetworks

When calculating the number of available hosts on a subnet, always subtract 2 (for the subnet ID and the broadcast)

You do not need to subtract 2 when calculating available subnets

subnets: 2^n

hosts in a subnet: $2^n - 2$



Review (cont'd)

The delta method for subnetting is a convenient way to calculate the number of bits you must move a mask to obtain the number of desired subnets

Each time you divide a network in half, you make its subnet mask one bit longer

The high level steps of the delta subnetting method are:

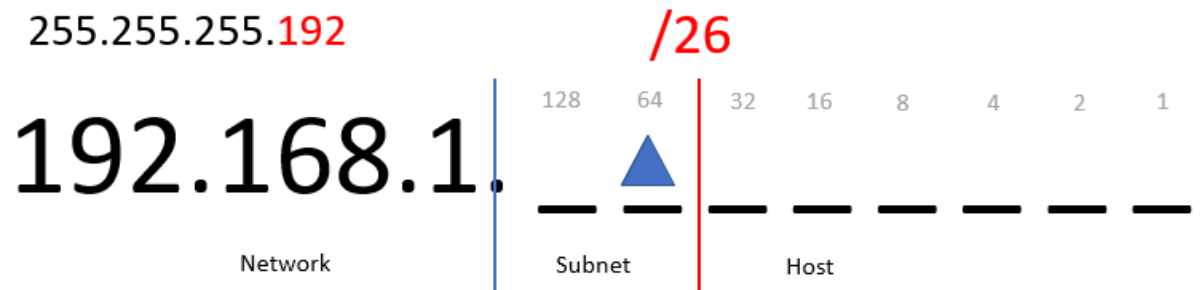
Identify the starting subnet mask

Determine # of bits to move the mask

Move the mask and identify the delta

Use the delta to calculate subnet IDs and IP ranges

When determining subnet IDs, increment by the delta



Review (cont'd)

IPv6 was created to resolve the eventual depletion of IPv4 public addresses

2601:140a:8780:43f0:2d1e:9ebb:92f0:d6e9, and fe80::10:18ff:fe12:2001 are examples of IPv6 addresses

IPv6 addresses can be configured manually, self-assigned, calculated via SLAAC and EUI-64, or obtained via DHCP

With SLAAC, the network half of the IPv6 address is obtained from the local router

EUI-64 modifies the MAC address to create a unique 64-bit number

IPv6 uses multicasts, but not broadcasts

You can simplify an IPv6 address by compressing contiguous 0s (represented as ::), or by leaving off leading 0s in a hextet

The IPv4 to IPv6 transition mechanisms are: dual stack, 6-4 tunneling, and 6-to-4 NAT

